# Housing First, Children Later*

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**Abstract**

Should housing assistance be conditioned on sobriety? With no externalities or distortions, and forward looking households accounting for the true hazard, conditionality simply removes an option, so Housing First weakly dominates Treatment First at equal generosity and cost. The case must then rest elsewhere. I separate a genuine externality, the distortions that fund the programme, and its own distortions, in a general equilibrium life-cycle model with endogenous homelessness, drug use and mortality, calibrated to United States data. The decisive channel is intergenerational: a parent's disorder and homelessness pass a worse draw to the child, and pricing it reverses the ranking.

*Keywords:* homelessness; housing vouchers; substance use disorder; intergenerational transmission; externalities.

*JEL classification:* I18; I38; H23; R21; J62.

## 1 Introduction

On a single night in January 2024 the point-in-time (PiT) count run by the Department of Housing and Urban Development (HUD) found 771,480 people homeless in the United States, the most since the count began. In July 2025 an executive order moved the federal government decisively to one side of a long argument about what to do about it: it required sobriety in federally funded housing and homelessness programmes and made it easier for states to commit people to treatment against their will.[1] The money followed, redirecting about a third of what the two programmes carry between them away from Housing First and towards programmes conditional on sobriety.[2] Whether housing assistance should be conditioned on sobriety is no longer an academic question. It is policy, and it decides the welfare of hundreds of thousands of people.

The argument between the Housing First and Treatment First camps is usually framed as a contest between two behavioural claims.[3] Critics of Housing First say that unconditional assistance is moral hazard: guaranteeing housing whatever a household does weakens the incentive to stay sober and attached to the labour market, so use and substance use disorder (SUD) rise, which feeds back to homelessness, because a SUD costs jobs and a lost job means not only more use but less ability to pay rent. Critics of Treatment First say that conditionality delays stable housing for exactly the households least able to wait, worsens their employment, and makes SUD recovery harder, because the street, or a shelter, is a poor place to recover. That too runs in a circle: worse recovery means more disorder, and more disorder means more homelessness. Both claims are true about behaviour, and the experimental literature has not settled between them, because the trials that could are short.

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[1] Executive Order, *Ending Crime and Disorder on America's Streets*, 24 July 2025. Reported in `https://www.npr.org/2025/07/24/nx-s1-5479139/trump-homelessness-executive-order-civil-commitment-camping`.

[2] See `https://www.npr.org/2025/11/14/nx-s1-5553561/homelessness-housing-funding-trump-administration-hud`.

[3] For the case against Housing First as the federal government now makes it, see the executive order cited above; for the case for it, see the National Alliance to End Homelessness, `https://endhomelessness.org/blog/ceo-corner-week-of-july-28/`.

Randomised trials of Housing First find large and persistent gains in housing stability and mixed to null effects on substance use (Tsemberis et al., 2004; Padgett et al., 2006; Larimer et al., 2009; Goering et al., 2014; Aubry et al., 2016), but they run six to twenty four months, so they cannot speak to long run behaviour or to the budget required to fund either policy at scale.

This paper starts from the observation that the debate as framed is missing the mark, because the moral hazard both sides argue about has nothing to do with efficiency at all. Consider households that are forward looking, as in Becker and Murphy (1988), and that have no self-control problems of the kind modelled with hyperbolic discounting at the moment they decide whether to start using. Once a household is addicted the assumption is a different and stronger one, and nothing below rests on it. Suppose they account for the true hazard of acquiring a SUD when they make that decision, as they do in the model in this paper and as the evidence suggests (Lundborg and Lindgren, 2002; van Ours, 1995; Grossman and Chaloupka, 1998; Saffer and Chaloupka, 1999; Caulkins, 2001; Gallet, 2014; Greenwood et al., 2025). Then at the same generosity and the same cost, Housing First weakly dominates Treatment First for every household. The two policies differ only in whether the voucher is withheld in one state, so the household's budget set under Housing First is weakly larger everywhere and strictly larger somewhere, and a household under Housing First can reproduce its Treatment First outcome exactly by simply not using. Conditionality removes an option from an optimiser and gives nothing back.

The case for Treatment First therefore has to rest on something outside the household's own objective, and this paper focuses on three such things. The first is a genuine externality: a cost the household imposes on someone whose welfare it does not fully count. The second is the distortions that fund the programme, since a policy that costs more to run raises a distortionary labour income tax. The third is the programme's own distortions, and they work through the means test and the clawback rules: a household that earns more loses eligibility and has its transfer clawed back, so it faces an implicit marginal tax on top of the statutory one and has a reason to work less in order to qualify and to keep more of what it qualifies for.

The model captures all three. First, it captures the latter two through endogenous labour supply and a labour income tax that must clear the government's budget, so a policy that costs more is paid for by a distortion the model prices rather than by assumption. As well, the programme's design is matched to the design of the real one, so the implicit marginal tax the means test creates is the one households actually face.

Second, it captures the externalities, of which there are two. The first is intergenerational.

The SUD externality works as follows. A parent's SUD lowers the child's own earnings, and a lower earning child is then likelier to decide to use and consequently to acquire a SUD itself, which then means they are likelier to lose work, to end up on the street and to die young. Evidence shows that children of a parent with a disorder are two to four times as likely to acquire one themselves (McGovern et al., 2023), and the earnings loss is also visible in children exposed in utero to a Swedish experiment that sharply raised alcohol availability, who earn about 24 percent less as adults (Nilsson, 2017). This half favours Treatment First, since any use that conditionality deters is a chain it never starts.

Homelessness also creates an externality, and it runs through the same chain: a parent's homelessness lowers the child's earnings, and everything that follows from lower earnings follows again. The evidence supports this channel as well. Children displaced from Chicago's public housing earned 16 percent more a year as adults (Chyn, 2018), and a move to a lower poverty neighbourhood before age thirteen raises adult earnings by 31 percent (Chetty et al., 2016). This half favours Housing First, since the policy that houses more people is the one that stops more of these chains.

All three penalties are disciplined against data. The disorder penalty is set so that a child of a parent with a SUD is three times as likely to acquire one itself, the middle of the range McGovern et al. (2023) report; the homelessness penalty so that the child of a homeless parent is twice as likely to be homeless itself, the midpoint of the two measures in Bassuk et al. (1997), and the homelessness penalty is checked against an earnings estimate the calibration did not use. What makes these externalities is that a parent counts only a fraction $\omega$ of what it does to its child, so the remainder is external whenever $\omega < 1$. The weight is not assumed: it is calibrated inside the quantitative model, to how far drug use rises when altruism is switched off, so the fraction a parent does internalise is disciplined by behaviour. Either way, what is a private problem for the parent is a childhood for the child, and the child does not choose it.

The impact does not end with the child: through the child it ripples out to the whole lineage. The parent

values its own child at $\omega$, that child values the grandchild at $\omega$ again, so generation $k$ enters the parent's problem at $\omega^k$: what one decision costs is a line of descendants, and the share left external rises at every link.

The second externality runs through the aggregate rather than through the family. Households dislike homelessness around them whether or not it is their own child's, so a policy that leaves more people on the street imposes a cost on everyone, and that is what makes the homelessness rate a genuine fixed point problem rather than accounting. The paper prices it as a disamenity $\kappa_x$ per unit of the aggregate homelessness rate in flow utility, set from the contingent valuation in Loubière et al. (2020), where the Europeans willing to pay anything to end homelessness would pay 0.20 percent of average annual earnings a year.

Where the model is truly superior to a randomised trial is in what it can follow. I can trace the impact of a policy well beyond the households in the study group, to others in the economy and to their children, and well beyond the length of a trial. And I can price what the policy does to the government's budget and to the economy as a whole, which a trial holds fixed by construction.

One mechanism of the model decides who the two policies treat differently, and it is central enough to state here. Two facts about drug use have to be reproduced together. It is concentrated at the bottom of the income distribution, and it is not universal: most households do not use at all, so there is a margin at zero, with a participation rate of 5.13 percent among adults aged 26 and over (NSDUH 2024). To generate both I model drugs as a luxury good bought at a price, which puts most households at a corner where they optimally consume none. Left to its own devices, however, that same assumption concentrates use at the *top*, since the share of income spent on a luxury rises with income. What reverses this income gradient, as I show, is the increased mortality risk that comes with a SUD, which I model and discipline with the data: acquiring a disorder truncates the expected continuation value of life, and that loss rises faster than income does, so in equilibrium use concentrates at the bottom.

I make the argument in two models, and I start by building the intuition. A stylized two period model, in Section 3, shows how the increased mortality risk mechanism reverses the income gradient at its clearest, and how the externality that overturns the dominance result works.

A full-fledged quantitative model follows, to quantify the mechanisms and the welfare implications of Housing First against Treatment First. It is a general equilibrium life cycle economy, in Section 4, with six state variables and fifty three parameters. It carries endogenous homelessness, drug use, the SUD, recovery, serious mental illness (SMI), employment and mortality, together with a drug market that clears, so drug prices respond to policy, and a government budget that balances. It also considers the costs of enforcing conditionality, and imperfect detection. Conditionality there is enforced by a test that is wrong in both directions, so some households lose the voucher though they meet the condition and some keep it though they do not, and the government pays to run the test. Both models are calibrated to United States data, and the calibration is set out parameter by parameter in Section 3.3 for the stylized model and Section 5 for the quantitative one.

The stylized model delivers the result in a form that can be read off. At universal cover and with no externality, 4.8 percent of households are strictly worse off under Treatment First and none is better off, so the private cost of conditionality is 0.105 percent of consumption in the aggregate: small, because so few households are affected, and unambiguous in sign. Switching the externality on, conditionality prevents 2.492 percentage points of disorder cases, and preventing one case is worth 204.1 percent of consumption to one child. The product is 5.086 percent, which is 48.4 times the private loss. Housing First is better for every household taken one at a time, and worse once one counts the children of the households it fails to deter. The ranking survives every value of the transmission penalty considered except zero, where by construction there is no externality to price.

The quantitative model is what settles whether that ranking survives in an environment that fits the data, where the drug price responds to the number of users, the rent responds to voucher demand, the tax rate responds to the programme's cost, and the two policies can be separated on generosity as well as on conditionality. It is also what allows the short run and the long run to be distinguished, which matters for reading the experimental evidence: if housing outcomes respond immediately to voucher design while disorder adjusts slowly, short trials are well powered to detect the first and not the second whatever the long run effect turns out to be.

## 2 Related literature

The literatures this paper draws on are set out in the order the argument needs them. The first three supply the machinery: what the trials establish about housing policy, how economists model the decision to use substances, and what structural models of homelessness already deliver. The next two supply the motivation, which is where this paper departs from its predecessors: what a parent with a SUD, or a parent on the street, transmits to their child, and how much weight a parent puts on the child who bears it. The last two are what the quantitative model is built out of: the hazards, taken one at a time from clinical and epidemiological work, and the drug market that makes the price endogenous.

**Randomised trials of Housing First.** A large empirical literature evaluates programmes that provide rapid housing without conditioning on sobriety, and finds large and persistent improvements in housing stability and reductions in shelter and other public service use (Tsemberis et al., 2004; Padgett et al., 2006; Larimer et al., 2009; Goering et al., 2014; Aubry et al., 2016). Evidence on substance use is mixed: most studies find no statistically significant change, and some report modest reductions in alcohol related outcomes. These evaluations typically measure outcomes over six to twenty four months and rarely beyond three years, which is short against a recovery process that Vaillant (1995) documents as gradual and protracted. The model here offers a structural reading of that pattern rather than a dismissal of it: housing responds immediately to voucher design, while a SUD adjusts slowly, so a short trial is well powered for the first and not the second whatever the long run effect is. The deeper limitation of a trial is not power but scope. A trial randomises households within a fixed environment, and holds fixed exactly the things that decide the welfare comparison: the price of the drug, the distortions of the taxes that pay for a programme at scale, and, most importantly, the outcomes of children who are not in the sample.

**Economic models of drug use.** Modelling use as forward looking optimisation dates to Becker and Murphy (1988), and the assumption is empirically defensible at the margin where policy operates. Both users and non users hold risk perceptions close to the objective probabilities (Lundborg and Lindgren, 2002), and Greenwood et al. (2025), who allow perceived and true SUD risk to differ, find the model implied beliefs are close to the true risk. The price evidence points the same way and is taken up with the drug market below, since the elasticity of participation is what disciplines the size of the behavioural response to Treatment First and to Housing First. A separate literature emphasises temptation and dynamic inconsistency (Laibson, 1997; Gruber and Kőszegi, 2001), under which present biased households over consume relative to their own long run preferences and corrective policy has a rationale with no externality at all. That literature is complementary here rather than opposed, and it cuts in the same direction: a present biased household would welcome a sobriety requirement as a commitment device, so introducing it would strengthen the case for conditionality rather than weaken it.

Where this paper departs from that literature is in what generates the income gradient in use. Greenwood et al. (2025) build a macroeconomic model of the opioid crisis in which households differ by skill, and the concentration of use among the less skilled is carried in part by preference differences towards drugs across those groups. In contrast, in the model here no preference parameter varies with income: every household has the same weight on the drug, the same Stone-Geary endowment, the same hazard per unit used and the same mortality penalty. The utility function does not change with the state the household is in, or with the choices it makes. The gradient comes out of the mortality channel instead, because a SUD destroys a continuation value and the continuation value rises with permanent income, so the household with more to lose loses proportionally more than it gains from a drug that is cheap relative to its income.

**Structural models of homelessness.** The paper builds directly on İmrohoroğlu and Zhao (2022), who model homelessness as a rational response to income risk, mental health shocks, housing costs and housing policy. In that framework housing policy reaches homelessness through income loss and housing costs alone, so a sobriety contingent policy has no behavioural margin to act on in their model. Adding endogenous use, disorder and recovery, together with state dependent mortality, creates the margin: the same voucher that insures a household also changes what using costs it, and the disorder it may acquire then feeds back into employment, housing and survival. Abramson (2026) makes the complementary point on the other side of

the market, showing that eviction policy changes landlord screening and rents and thus changes who becomes homeless in equilibrium. Here the causation runs both ways, and the two directions reinforce each other. On the one hand, the SUD costs employment, which costs a household its housing. On the other hand, losing housing worsens employment and raises mortality, which raises the value of using relative to the value of life being risked, since a household with less continuation left to lose risks less by using drugs. Each direction feeds the other, so what the model produces is a vicious circle rather than a one directional flow.

**Intergenerational transmissions.** Three links have to hold for the argument to work, and each is estimated separately.

The first is that a parent's SUD changes the childhood. The administrative record of the child welfare system documents it at scale: parental substance abuse is recorded at more than a third of removals. The epidemiological literature documents it too. Household substance abuse is one of the adverse childhood experiences of Felitti et al. (1998), and children of a parent with a SUD are two to four times as likely to acquire one themselves (McGovern et al., 2023).

The second link is that the childhood shows up in the child's own adult earnings, and the mechanism is as follows. A parent who has lost their job to the SUD, and who spends what money is left on the drug, has less of it for the child. Time goes the same way. What the child loses is schooling and the ordinary attention a household gives, in a home more chaotic than it would otherwise be. The child reaches the labour market with less human capital for it.

Each step of that is measured, and each has been built into a model before. Dahl and Lochner (2012) find that family income raises children's achievement, using the Earned Income Tax Credit to move the income, and Restuccia and Urrutia (2004) put roughly half of the intergenerational correlation in earnings down to parental investment in education, most of it early. Card (1999) puts the return to a year of schooling at around ten percent. And Nilsson (2017) closes the loop on a parental substance shock in particular: children exposed in utero to a Swedish experiment that sharply raised alcohol availability earn about twenty four percent less as adults and finish less schooling, a magnitude of the same order as the penalty $\delta_d$ that this paper calibrates inside the model. What no study does is estimate the effect of a parent's SUD on a child's lifetime earnings directly. That is why $\delta_d$ is calibrated inside the model rather than lifted from a coefficient: these estimates imply that the effect is real and of first order, and the penalty itself is set to reproduce what the literature does measure for the children of parents with a SUD: they are two to four times as likely to acquire one themselves.

The third is that housing, and the poverty that comes with bad housing, is one of the environments doing the transmitting. Chyn (2018) finds that children displaced by Chicago's public housing demolitions earned sixteen percent more a year as adults. Bassuk et al. (1997) is what the homelessness penalty is calibrated to, and its case control design is what makes it usable: among 220 homeless mothers and 216 low income housed mothers, 14.2 against 8.3 percent had a primary female caretaker who had been homeless, and 10.3 against 4.4 percent a male one. Those are odds ratios of 1.80 and 2.41, and $\delta_h$ is set so the model reproduces their midpoint. Because both arms are poor, the comparison isolates transmission rather than the poverty that accompanies it, and no population base rate has to be assumed.

Chetty et al. (2016) then becomes a check rather than the target, and it is the stronger for it. A move to a lower poverty neighbourhood before age thirteen raises adult earnings by thirty one percent, and the gain falls with the age at the move, which says that what is transmitted is the environment a child is raised in rather than anything the child carries. The penalty the model needs to match Bassuk implies an earnings loss of 33 percent, so an estimate the calibration never saw reproduces one it did. Fogli and Guerrieri (2019) put the same mechanism inside a general equilibrium model, where parents choose the neighbourhood they raise a child in and local spillovers feed back into the return to education; $\delta_h$ is the reduced form of that feedback rather than a competing account of it.

What this paper adds is not another estimate but a price: it puts these magnitudes inside an equilibrium model where the parent's use decision is endogenous, so the transmitted loss becomes a quantity that policy moves rather than a coefficient.

**How much a parent weighs its child.** The externality is whatever the parent does not internalise, so its size is decided by the weight a parent puts on its child, and that weight has a literature of its own. Altonji et al. (1997) test pure altruism directly, on consumption within parent and child households, and reject it: the transfers parents make to their children while alive do not respond to the distribution of income inside the family the way complete altruism requires. Parents are probably altruistic, but not completely, which is the case for treating $\omega$ as strictly between zero and one rather than assuming either end of it. Their experiment cannot be run here. It compares parents against their own adult children, and every household in this model has a child, born when it dies, so there is no parent against non parent comparison to make in the benchmark. What the model can do is another test, and for this paper it is a more relevant one: run the comparison as a counterfactual. In NSDUH 2024 past month use among parents is 0.845 of use among non parents, reweighted to the non parents' joint distribution of age, income, sex and marital status so that the gap is behaviour rather than composition. The child term of the household's problem is the only thing in the model that makes a household use less than it otherwise would, so $\omega$ is set so that switching that term off raises use by the same 18.3 percent. The externality is therefore not assumed to be large, and its size is calibrated to observed behaviour on exactly the margin this paper is about.

**What the hazards are built from.** The model's transitions are not free parameters dressed as structure; each is anchored on a clinical or epidemiological estimate, and the calibration section names them one at a time. Four groups do the work. On mortality, Hjorthøj et al. (2023) put the excess death rate for a SUD at six to ten times the baseline, Baggett et al. (2013) measure it for the homeless by age, and Olfson et al. (2015) do the same for serious mental illness; against those hazards the value of a life comes from the willingness to pay estimates surveyed in Viscusi and Aldy (2003). On the mental illness chain, Nielsen et al. (2017) identify the effect of a SUD on the onset of SMI in Danish registers, Gutwinski et al. (2021) meta analyse its prevalence among the homeless, and Bradizza et al. (2006) review relapse under dual diagnosis. On recovery, Vaillant (1995) documents how slow unassisted remission is, and Milby et al. (2005) supply the causal estimate this paper leans on hardest: a randomised trial on cocaine dependent homeless adults in which the housed arm was abstinent 6.60 weeks against 4.48 in the unhoused arm, which is where both the treatment effect on recovery and the penalty homelessness imposes on it come from. On employment and the cost of housing loss, Desmond and Gershenson (2016) estimate what losing housing does to holding a job, and Loubière et al. (2020) price the disamenity of visible homelessness through a contingent valuation, which is what makes the aggregate headcount enter anybody's utility at all. Taking these from outside the model is the point: the parameters this paper argues over are the ones in Section 5, and a reader who disagrees with one of them is disagreeing with the study behind it.

**The drug market.** Making the drug price endogenous is what lets a policy that changes the number of users change what using costs everybody else. Two elasticities govern it and they do different jobs, so the literatures behind them are worth separating: the demand side is driven by how much the extensive and the intensive margins respond to a given price, and the supply side determines how much the price moves when demand changes.

*The elasticity of participation.* This is the extensive margin of the demand side, and the most relevant margin for the comparison this paper makes, since Treatment First and Housing First differ in who is drawn into use at all. It is what fixes how large the behavioural response to either policy can be in the model, and the model is checked against it in Section 5.3 rather than calibrated to it. The intensive margin, how much a user buys once using, is a separate elasticity that the model produces separately. Gallet (2014) synthesises 462 price elasticity studies, and van Ours (1995), Saffer and Chaloupka (1999) and Caulkins (2001) establish that participation responds to price even for drugs as habit forming as heroin and cocaine, with long run responses consistently larger than short run ones. That ordering is the signature of forward looking behaviour rather than myopia, which is why it belongs with Becker and Murphy (1988) above as well as here.

*The elasticity of supply.* This is the general equilibrium offset, and it runs against Treatment First in particular: curing households takes them out of the market, demand and the drug price fall, and a cheaper drug draws others in. I am not aware of a direct estimate for an illicit drug. Cannabis is the closest market I found to an illicit drug, as it remains a Schedule I substance federally, so even a licensed producer works

without interstate trade, without ordinary banking and under federal prohibition, which is the set of frictions an illicit supplier faces. Mace et al. (2020) estimate a price elasticity of supply of 1.46 on Washington State administrative data, and that is the number the model uses.

**In a nutshell.** The trials establish that housing stability responds to policy but cannot price what happens in the long run, to others not in the sample, or to the distortionary tax system that pays for it. The rational addiction literature models the use and SUD decisions but abstracts from housing, children, and mental illness. The structural homelessness literature delivers a housing market equilibrium but has no drug margin for conditionality to act on. The transmission literature measures what a childhood costs but takes the parent's behaviour as given. This paper's contribution is to put them in one framework and then to insist on a distinction none of them needs on its own: that the moral hazard which dominates the policy debate is not a reason for conditionality, and that whatever case exists has to be made on the externality, on the tax, or on the design, one at a time and with the three kept apart.

# 3 A stylized model

## 3.1 Three questions

In the data, drug use is concentrated among low income households, most of the homeless do not carry a SUD, and housing vouchers reach only about a quarter of the households eligible for them. The quantitative model is built to match facts of that kind. It has six state variables and fifty four parameters, which is what the welfare and fiscal questions require, but it also makes it hard to see which pieces of the environment do the heavy lifting. This note strips the environment down to help understand the main drivers in the model, but where my three main questions are still well posed.

Households are perfectly forward looking, here and in the quantitative model. They face the true hazard of acquiring a SUD, they know what carrying one would cost them, and they make no mistakes. Nothing below rests on misperception, on myopia, or on a taste that changes once the household has used.

1. **Who uses drugs?** In the data, use is concentrated at the bottom of the income distribution. Drugs here are a *luxury* good bought at a price: the Stone-Geary form in equation (1) below means the share of income spent on them rises with income, so left to itself the environment concentrates use at the *top*. Which feature of the environment reverses the concentration from the top to the bottom?
2. **How does housing assistance shape the drug use decision?** A voucher pulls drug use in two opposite directions. It raises the value of staying clean, because a housed future is worth more than a homeless one and using puts that future at risk: call this the value of life force, and it lowers use. In contrast, it cushions the consequence of acquiring a SUD, because a household that acquires one is housed anyway: call this the moral hazard force, and it raises use. Both are first order, so theory does not sign the effect of housing assistance on drug use, and the question is quantitative.
3. **When is Treatment First better than Housing First?** In a model without externalities, and where the two programmes cost the same to run, never. Housing First always raises welfare when agents are forward looking, as they are here. A SUD creates a genuine tension, since the same voucher that insures a household also makes the risk it insures more attractive, but that tension does not raise welfare, because Treatment First only removes an option that a household can take under Housing First as well. However, if a SUD and homelessness generate externalities, such as affecting the earning potential of the children of parents with a SUD, then Treatment First may be preferred: children of a parent with a SUD are themselves two to four times as likely to develop one (McGovern et al., 2023), and moving a child to a lower poverty neighbourhood before age thirteen raises adult earnings by 31 percent (Chetty et al., 2016). The model's work here is to show why Housing First raises welfare whenever there are no externalities (and no distortions from the programmes or from the taxes that pay for them, both of which the full model takes up), and then show how the externality that overturns it works.

The model is qualitative throughout: the calibration is rough by design, there is no validation exercise, and every component that obscures the mechanisms has been removed and the model refitted without it. What remains are the main drivers of the results. The welfare evaluations below are for illustration only; the full quantitative model is what settles the main question, which is whether Treatment First raises welfare relative to Housing First in a model that fits the data.

## 3.2 The environment

### 3.2.1 Households and preferences

There are two periods and a continuum of households. A period stands for thirty years of adult life, so the two of them span ages twenty five to eighty five. Parameters with a natural annual counterpart are compounded to that length: the discount factor is $\beta$ per period, the gross interest rate is $R = 1/\beta$ so that $\beta R = 1$, and $\phi_0$ is survival over thirty years. Notation follows the quantitative model throughout.

A household is characterised by its productivity $y$, fixed over its life and lognormally distributed across households, $\log y \sim \mathcal{N}(-\frac{1}{2}\sigma_y^2, \sigma_y^2)$, so mean income is one. Households differ in $y$, in age $j \in \{1, 2\}$, in employment status $e \in \{0, 1\}$, in voucher receipt $\eta \in \{0, 1\}$ and in SUD status $d \in \{0, 1\}$, all discussed below, and in nothing else.

Flow utility depends on non durable consumption $c$, housing services $h$ and drug consumption $v$,

$$u(c, h, v) \;=\; (1 - \gamma - \theta)\log c \;+\; \gamma \log h \;+\; \theta\big[\log(b + v) - \log b\big] \;+\; \aleph. \tag{1}$$

The three weights sum to one, so $\gamma$ is the housing budget share. Drugs enter only in the first period, since that is the only period in which $v$ is chosen, and housing is chosen each period from a three point grid $\{h_1, h_2, h_3\}$. The pair $(\theta, b)$ governs the drug term: the marginal utility of the $v$th unit is $\theta/(b + v)$, so $\theta$ mostly sets how much a user uses and $b$ whether anyone uses at all. The Stone-Geary form keeps the first unit's marginal utility finite, which is what generates non users at all, because under plain $\theta \log v$ every household uses whatever the consequences. Finally $\aleph$ is the flow value of being alive, the utility of surviving into a period over and above what is consumed in it, which gives mortality a cost that does not vanish when consumption is low. Households save in one risk free asset and cannot borrow.

### 3.2.2 Timing

**Period 1.** Every household is clean. Employment $e \in \{0, 1\}$ is drawn with the unemployment probability $u_0$, and a household that is income eligible draws a voucher with probability $\pi_V$; these are the same two lotteries it faces again in period 2. It then chooses consumption, housing, drug use $v \geq 0$, which for simplicity is the only period in which it chooses whether to use, and saving, paying market rent $p_r$, which is normalised to one, on whatever unit it picks. It starts with no assets. Using drugs generates the risk of a SUD, increasing in $v$,

$$\pi(v) \;=\; 1 - e^{-\lambda v}. \tag{2}$$

Because $\pi$ is increasing in $v$, the household internalises that using more is riskier, so the intensive and extensive margins are not separate decisions.

**The transition.** The SUD $d' \in \{0, 1\}$ is realised with probability $\pi(v)$ for households that used and with probability zero for the rest, and the household survives into period 2 with probability

$$\phi(d') \;=\; \phi_0\,(1 - \kappa d'), \tag{3}$$

so $\kappa$ is the proportional reduction in survival caused by the SUD. The mortality penalty applies here, and here only.

**Period 2.** Terminal, after which the household dies with certainty. A household without the SUD is unemployed with probability $u_0$ and one with the SUD with the higher probability $u_1$; if unemployed it receives income support. The voucher is drawn again, subject to the policy rule. The household chooses consumption and housing, there is no drug use decision, and assets are run down with nothing bequeathed.

**Mortality, and where the SUD is realised.** Mortality is a tax on a continuation value, so the SUD can only cost anything if a future remains when it is carried. That does not require a third period. It requires the SUD to be drawn at a *transition* rather than inside the terminal period, and here it is drawn on the way into period 2, with survival over that transition depending on it, so acquiring it puts the whole of period 2 at risk. Realise it inside the last period instead and death costs nothing because nothing follows it, and the mechanism the model rests on falls silent.

### 3.2.3 Employment, income support and housing

The gap $u_1 - u_0$ is the entire resource cost of the SUD. Income in a period, given permanent productivity $y$ and employment status $e$, is

$$y^g(y,e) \;=\; \begin{cases} y & \text{if } e = 1, \\ \psi \cdot \min\{y, \bar{y}\} & \text{if } e = 0, \end{cases} \tag{4}$$

with $\psi$ the replacement rate and $\bar{y}$ a cap on the earnings base. There is no other income risk. The level of $\psi$ is what links the SUD to homelessness: for the poorest households half of income can fall short of the rent on the smallest unit, so an unemployed household there cannot house itself out of current income.

Each period the household chooses

$$h_k \in \{h_1,\ h_2,\ h_3\}, \qquad h_1 < h_2 < h_3, \tag{5}$$

homelessness ($h_1$), a small unit ($h_2$) and a standard unit ($h_3$). Homelessness is free and delivers less housing services than the market does.[4]

A voucher covers part of the rent, is means tested and is rationed. It is drawn independently each period, which is a simplification relative to the quantitative model, where a voucher persists as long as eligibility holds. Writing $\eta \in \{0,1\}$ for whether one is held,

$$\Pr(\eta' = 1 \mid y', d', v) \;=\; \pi_V \cdot \mathbf{1}\{y' \le y_V\} \cdot \mathbf{1}\{\text{policy covers } (d', v)\}, \tag{6}$$

and a household holding one pays the fraction $\kappa_r$ of market rent, so the rent paid on option $k$ is

$$r_k(\eta) \;=\; p_r\, h_k \big[\kappa_r \eta + (1 - \eta)\big] \cdot \mathbf{1}\{k \neq 0\}, \tag{7}$$

which is zero when the household is homeless. Vouchers operate in both periods, and $\pi_V$, the share of eligible households actually covered, is the policy dial of Section 3.5.

The two policies differ in exactly one respect, the last indicator in equation (6). **Housing First** covers every eligible household regardless of $(d', v)$. **Treatment First** covers only households that satisfy the sobriety condition, $v = 0$ and $d' = 0$, which is the rule in the quantitative model as well. Since policy coverage in the states that follow drug use is the only difference, the gap between them is the moral hazard and the whole of it.

The condition binds in period 1 as well, and the timing there is worth stating. Every household arrives clean, so the period 1 voucher is drawn before $v$ is chosen and conditionality cannot screen anyone out on entry. The order within the period is therefore: the household chooses $v$, the policy rule is then applied to the voucher it drew, and it chooses housing knowing what rent it faces. So under Treatment First a household that uses would be subject to market rent immediately and again in period 2.

### 3.2.4 The household's problem

Let $a$ be assets carried in. Period 2 is terminal, so the household consumes everything and chooses housing,

$$W_2(a, y, e, \eta, d) = \max_k \Big\{ (1 - \gamma - \theta) \log c + \gamma \log h_k + \aleph \Big\}, \qquad c = Ra + y^g(y,e) - r_k(\eta), \tag{8}$$

[4] Two rented sizes is the fewest that lets a household economise on housing rather than lose it. With one, homelessness comes out at 19.52 percent against a target of 0.40, at any $h_1$.

and write $W_2(a, y, d) = \mathbb{E}\big[W_2(a, y, e, \eta, d) \mid y, d\big]$ for its expectation over the period 2 employment and voucher draws. In period 1 the household draws $(e, \eta)$ and then chooses housing, drug use and saving,

$$V_1(y, e, \eta) = \max_{k,\, v \geq 0,\, a' \geq 0} \Big\{ (1-\gamma-\theta) \log c + \gamma \log h_k + \theta[\log(b+v) - \log b] + \aleph + \beta\, \mathbb{E}_{d'} \Big[ \phi(d')\, W_2(a', y, d') \Big] \Big\}, \quad (9)$$

with $c = y^g(y, e) - r_k(\eta) - p_v v - a'$ and $d' = 1$ with the use dependent probability of (2). Survival sits inside the expectation over $d'$ because it depends on which $d'$ is realised.

A household uses if the value at its best positive $v$ exceeds the value at $v = 0$. Households differ only in $y$ and in the draw $(e, \eta)$, which is the same lottery for everyone, so within a draw the decision defines a set of income intervals on which use occurs, and participation at a given income is the probability of drawing a state in which the household uses.

## 3.3 Calibration

### 3.3.1 External calibration

Table 1 lists the parameters set outside the model. Preferences and demographics are conventional: the discount factor over a thirty year period is $\beta = 0.860$[5] and the gross interest rate is $R = 1/\beta$, and survival over a period is $\phi_0 = 0.91$, from the Social Security Administration (SSA) period life table for twenty five to fifty five. For the SUD, the proportional fall in survival is $\kappa = 0.49$ , derived by compounding an annual mortality ratio of 8, the midpoint of the range in Hjorthøj et al. (2023), so a clean household's chance of dying before period 2 is 9 percent and a household that acquires a SUD has a 53 percent chance of dying before period 2.[6] The two unemployment probabilities are measured directly. $u_0 = 0.049$ is the headline civilian unemployment rate from the Bureau of Labor Statistics (BLS) Current Population Survey, series UNRATE as distributed by Federal Reserve Economic Data (FRED), averaged over 2014 to 2024. And $u_1 = 0.1625$ is from the National Survey on Drug Use and Health (NSDUH) 2024 for respondents aged 26 and over.

Income support replaces half of previous earnings on a base capped at mean income, in line with statutory state rules. The two rental sizes are set such that the standard unit costs 0.41 of mean income and the small unit is a quarter of it. The voucher parameters are institutional: covered tenants pay 30 percent of market rent, eligibility is capped at mean income, and $\pi_V = 0.27$ is the assisted eligible share in the 2023 Picture of Subsidized Households, published by the Department of Housing and Urban Development (HUD). The weight on drugs is set at $\theta = 0.20$.[7]

### 3.3.2 Internal calibration

Six parameters are set inside the model, in Table 2, each bracketed on the moment it moves most rather than fitted jointly. The dispersion of log permanent income $\sigma_y$ is matched to how unequal earnings are at the bottom: in the 2022 Survey of Consumer Finances, households aged 26 to 65 with positive wage and salary income hold 17.2 percent of total earnings in the poorer half of the distribution. The share falls in $\sigma_y$, so one number identifies it, and the quantitative model of Section 4 is calibrated to the same moment. The weight on housing $\gamma$ is matched to the 26.8 percent of total expenditure that shelter plus utilities take in the 2024 Consumer Expenditure Survey (CEX), housing over expenditure net of drugs. The flow value of being alive $\aleph$ is matched to a value of a statistical life of 80 times average annual income, about \$9 million

[5] The annual discount rate here is higher than the one in the quantitative model, and this is deliberate. A period here is thirty years, so every consequence of using drugs arrives thirty years after the decision, whereas in the quantitative model consequences arrive annually and compound. A low annual discount factor would weigh that distant death much less and the mortality channel would be losing to impatience rather than to anything the model is about, so a higher factor corrects for the coarseness of the horizon rather than making a claim about preferences.

[6] The second figure is not $\kappa$ and should not be read as one: $\kappa$ is the proportional fall in the chance of *surviving*, so survival goes from $\phi_0 = 0.91$ to $\phi_0(1-\kappa) = 0.47$ and the chance of dying is one minus that.

[7] A $\theta$ large enough to match the 34.7 percent of income the average user spends on drugs makes the drug too valuable to be deterred by a death that arrives thirty years later, and users stop being poor: by $\theta = 0.23$ the mortality channel no longer reverses the income gradient on its own, and the average user is richer than average. Setting $\theta = 0.20$ gives a spending share of 19.34 percent and keeps the gradient, which is the trade this very simple model can make.

Table 1: Externally set parameters

| Parameter | Description | Source or rationale | Value |
|---|---|---|---|
| *Preferences and demographics* | | | |
| $\beta$ | Discount factor over a thirty year period | 0.995 annually, compounded | 0.860 |
| $R$ | Gross interest rate over a thirty year period | Set to $1/\beta$ | 1.162 |
| $\phi_0$ | Survival over a period, no SUD | SSA period life table, ages 25–55 | 0.91 |
| $\theta$ | Weight on drug use | Set; see the text | 0.20 |
| *The SUD* | | | |
| $\kappa$ | Fall in survival with a SUD, over a thirty year period | Hjorthøj et al. (2023), age 26+ | 0.49 |
| $u_0$ | Unemployment probability, no SUD | BLS CPS via FRED, 2014 to 2024 | 0.0485 |
| $u_1$ | Unemployment probability, with a SUD | NSDUH 2024, age 26+ | 0.1625 |
| *Income support* | | | |
| $\psi$ | Replacement rate while unemployed | DOL, statutory UI rates | 0.50 |
| $\bar{y}$ | Cap on the earnings base, multiple of average income | DOL, state benefit maxima | 1.00 |
| *Housing, drugs and the voucher* | | | |
| $p_r$ | Rent per unit of housing services | Normalisation | 1.00 |
| $p_v$ | Price of a unit of drugs | Normalisation | 1.00 |
| $h_3$ | Standard rental unit | Set | 0.44 |
| $h_2$ | Small rental unit | Set | $0.25 \times h_3$ |
| $\kappa_r$ | Tenant's share of the rent when covered | HUD voucher rules | 0.30 |
| $y_V$ | Voucher means test, multiple of average income | HUD income limits | 1.00 |
| $\pi_V$ | Voucher generosity in the baseline | HUD 2023, assisted eligible share | 0.27 |

Note: every parameter here is set from outside, so none has a model counterpart to report against the data. This is the same distinction the Model column of Tables 5 and 6 makes with n.a., which those tables need because they carry externally set and internally calibrated parameters together.

Table 2: Internally calibrated parameters

| Parameter | Description | Target moment | Data | Model | Value |
|---|---|---|---|---|---|
| $\sigma_y$ | Dispersion of log permanent income | Bottom half's share of earnings, SCF 2022 | 17.20% | 17.20% | 1.147 |
| $\gamma$ | Weight on housing services | Housing over non drug spending | 26.80% | 27.43% | 0.366 |
| $\aleph$ | Flow value of being alive | Value of a statistical life | 80.0 | 80.0 | 2.638 |
| $b$ | Pull of the first unit of drugs | Drug participation rate | 5.13% | 5.13% | 0.00272 |
| $h_1$ | Housing services when homeless | Homelessness rate | 0.40% | 0.35% | 0.00011 |
| $\lambda$ | Addiction hazard per unit of use | Adult SUD rate, NSDUH 2024 | 3.68% | 3.67% | 253.1 |

Note: the homelessness rate comes out at 0.35 percent against a target of 0.40, and the gap is left rather than closed. Every parameter after $h_1$ in the sweep moves homelessness a little: $h_1$ is bracketed on the homelessness rate, but raising it also raises participation, and $b$ is bracketed on participation afterwards, which pulls homelessness back down. Chasing the last tenth of a percentage point would mean alternating the two to a fixed point, which is machinery this model does not need. It is a model for intuition, and nothing below turns on the homelessness rate being exact.

against a mean household income of $113,000, which is the range surveyed by Viscusi and Aldy (2003) once their estimates are put in current dollars. The pull of the first unit $b$ is matched to a drug participation rate of 5.13 percent from NSDUH 2024 for respondents aged 26 and over, housing services on the street $h_1$ to a homelessness rate of 0.399 percent from the 2023 HUD point in time (PIT) count, and the SUD hazard $\lambda$ to the 3.68 percent of adults aged 26 and over carrying a SUD in NSDUH 2024, which is the moment the quantitative model matches too.

## 3.4 Who uses drugs

This section asks which feature of the model puts drug use at the bottom of the income distribution, and it starts from the force that any answer has to overcome. Drugs are a luxury good bought at a price, so a household with twice the income gives up less utility per dollar spent on them. With no consequences at all for acquiring a SUD that is the only force operating, and drug use concentrates at the top: the average user then earns 4.15 times the population average.

Two features of the environment are candidates to reverse it. The *mortality channel* is that the SUD

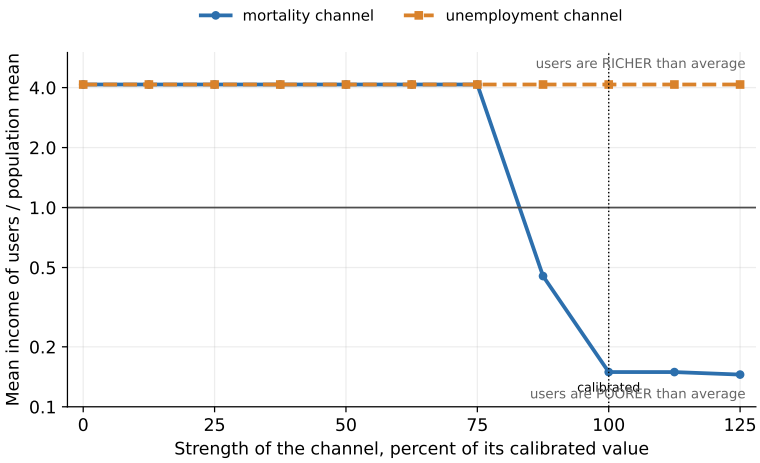



Figure 1: What it takes to reverse the income gradient in drug use

Note: the vertical axis is average income of users over the population average, on a log scale, and $b$ is recalibrated at every point so that participation is 5.13 percent throughout. The black dotted line marks the calibrated strength; everything to the right of it is extrapolation.

increases the chance of dying, through $\kappa$ in equation (3). The *unemployment channel* is that it costs you a job with probability $u_1 - u_0$, and if income support cannot cover the rent you downsize, live on less, or lose your home. Figure 1 scales each from nothing to 125 percent of its calibrated value, recalibrating $b$ at every point so participation stays at 5.13 percent. The amount of drug use is therefore identical all along both curves and the only thing moving is which households use.

The result is sharper than the statement that mortality matters more. That channel takes the income ratio from 4.15 with nothing switched on down to 0.15 at full strength, and it does so as a threshold rather than gradually: at half of its calibrated value the ratio is unchanged at 4.15, and the whole reversal happens over the last quarter of the channel. That threshold is a feature of the mechanism rather than an artefact. What mortality destroys is a continuation value, and a household only stops using when the expected loss of that value exceeds what the drug is worth to it; below the point where it does, a stronger penalty changes nothing about who uses, and above it the ordering flips at once. Past the calibrated point the ratio is flat, 0.15 at 125 percent against 0.15 at 100, so the calibration does not sit on the edge of the cliff but on the floor beyond it. The unemployment channel does not reverse it at all, leaving users at 4.15 times average income even fully switched on, and at 125 percent it is 4.15, still above one and no closer to turning. Strengthening the resource penalty by a quarter does not begin to do what mortality does. A resource penalty does not produce the concentration of drug use at the bottom.

A mortality penalty is different in kind. A resource penalty scales with the very thing it is meant to offset, since taking a fixed share of a large income still leaves a large income, so it cannot outrun the fact that drugs are cheaper in utility terms for the rich. What mortality destroys is not resources but a continuation value, and the continuation value rises with permanent income. The household with more to lose loses more, and loses proportionally more than it gains from drugs being relatively cheap. That is the work of $\aleph$ in equation (1): it is the part of the continuation value that does not shrink as consumption falls.

## 3.5 How does housing assistance shape the drug use decision?

The two forces of question 2 can be separated exactly, because they differ in what happens to a household that uses. Under Treatment First the voucher is withdrawn from any household that violates the sobriety condition, so only the value of life force operates. Under Housing First it is not, so both channels operate. Comparing the two policies at the same generosity therefore isolates the moral hazard force, and comparing Treatment First with no programme at all isolates the value of life force.

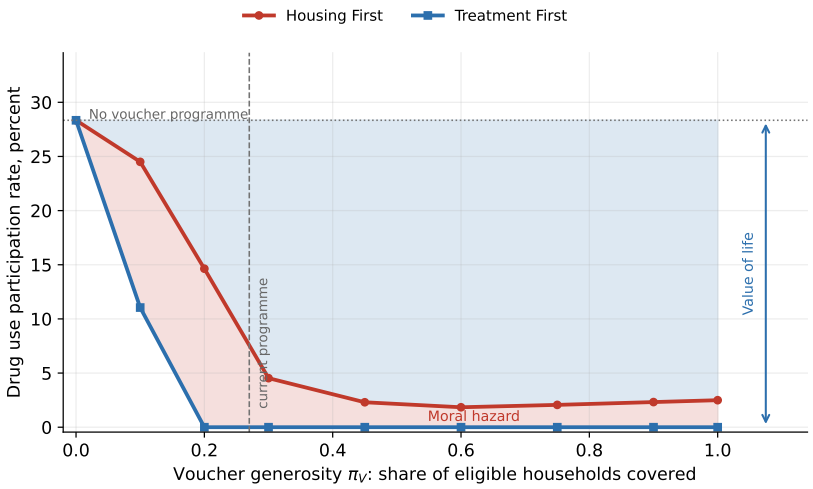



Figure 2: Drug use under the two policies as the voucher becomes more generous

Note: the red band is the moral hazard force and the blue band above it is what Housing First delivers net; together they are the value of life force, marked at the right. The dashed line is the generosity of the current programme, $\pi_V = 0.27$.

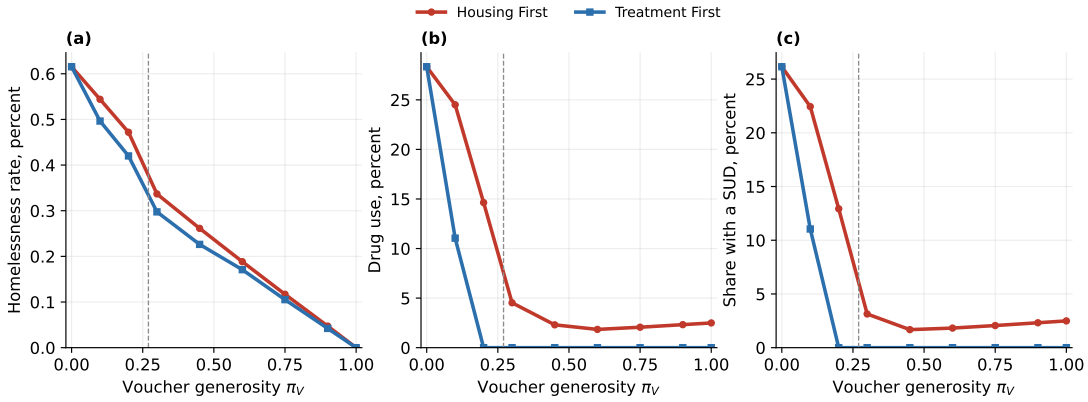



Figure 3: Homelessness, drug use and the SUD under the two policies

Figure 2 traces both as generosity rises. With no programme at all, 28.34 percent of households use. Universal Treatment First takes that to 0.00 percent, a value of life force of -28.34 percentage points; universal Housing First leaves it at 2.49 percent, so the moral hazard force is +2.49 points and the net effect of Housing First is -25.84. Both forces are first order and they work against each other, which is the central feature of the problem. Anyone asserting that housing the homeless must increase drug use, or must reduce it, is making a quantitative claim that needs to be tested.

Figure 3 plots the homelessness and SUD outcomes for each programme as its generosity changes, alongside drug use. In panel (a) the two policies end in the same place on homelessness, both driving it from 0.62 percent with no programme to zero, and along the way Treatment First is if anything slightly below Housing First, because the households it deters from using never acquire the SUD that puts their housing at risk. Panel (b) repeats Figure 2 here for convenience. The larger difference is in panel (c), where at universal cover Housing First leaves 2.492 percent of the population carrying the SUD and Treatment First leaves 0.000 percent. What the moral hazard buys is extra SUD cases among households that are housed either way.

Table 3: Welfare at universal cover, with no externality

| | Housing First | Treatment First |
|---|---|---|
| *Panel A. Outcomes at universal cover* | | |
| Drug participation rate | 2.49% | 0.00% |
| Share with a SUD | 2.492% | 0.000% |
| Homelessness rate | 0.000% | 0.000% |
| Voucher outlay per household | 0.3338 | 0.3350 |
| *Panel B. Treatment First against Housing First, no externality* | | |
| Households strictly worse off | | 4.8% |
| Households strictly better off | | 0.0% |
| Consumption equivalent, those affected | | 2.19% |
| Consumption equivalent, whole population | | 0.105% |

Note: the consumption equivalent is the proportional change in consumption, in every period and state, worth as much as the difference in lifetime value. Voucher outlay is expected spending on a household over both periods, the second discounted and weighted by survival, per unit of average income.

## 3.6 Welfare

### 3.6.1 Without externalities, Housing First always wins

Sections 3.4 and 3.5 establish what each policy does to behaviour. This section asks what each does to welfare, comparing the two at the same generosity $\pi_V$, whose outcomes are in Panel A of Table 3. The policies differ only in the last indicator in equation (6), so Housing First transfers resources in a superset of the states in which Treatment First does. The budget set is weakly larger in every state and strictly larger in one, so a household that optimises cannot be worse off under Housing First:

$$V_1(y;\ \mathrm{HF},\ \pi_V)\ \geq\ V_1(y;\ \mathrm{TF},\ \pi_V) \qquad \text{for every } y, \tag{10}$$

with equality exactly for households that never use. Put the other way round, a household under Housing First can reproduce its Treatment First outcome exactly by not using: conditionality removes an option from a forward looking optimiser and gives nothing back. This is a dominance argument, not a calibrated result, and no choice of parameters overturns it.[8]

It follows that the moral hazard is not an inefficiency. Households here face the true SUD hazard and make no mistakes, so the extra use Housing First induces is not a wedge to be closed but a household optimally taking a risk it is now better able to bear. The +2.49 points of Section 3.5 are a statement about drug use and not about welfare. Panel B of Table 3 puts numbers on it: 4.8 percent of households are strictly worse off under Treatment First, none is better off, and those affected lose 2.19 percent of consumption. Those are exactly the households that use: a household that never uses faces the same voucher rule under both policies and is indifferent between them. The figure is not the 5.13 percent participation rate of Section 3.3 because the programmes compared here are universal, rather than the rationed benchmark that rate is measured at.

A fixed budget does not rescue the welfare case for conditionality either. Treatment First is slightly *more* expensive than Housing First at universal cover, 0.3350 against 0.3338, so matching on cost rather than on generosity changes nothing. Two things offset. Survival into period 2 falls from 0.91 to 0.47 with a SUD, so the extra households Housing First puts into the SUD state die sooner and fewer are alive in period 2 to collect a voucher; against that, Treatment First withdraws the voucher from anyone who used, so it pays for fewer tenants while they live. The first effect is the larger, so the conditional programme ends up marginally the dearer, and the gap is small enough that nothing turns on it. So a case for conditionality has to come from an externality, and, in this model, from nothing else: with the two outlays equal and financing outside the model, there is no distortion left to tell them apart.

[8] It does, however, take the price of the drug and the cost of the programme as given, both of which this model fixes and the quantitative model does not. There the tax rate that pays for the programme is endogenous, so a policy that costs more raises the tax on everybody, and the drug price moves with the number of users. Either can separate the two policies with no externality present at all, which is why the argument here is stated for this model and not asserted of the other one.

Table 4: The externality on children

| *Panel A. Transmission: what each income penalty implies* $\delta_d$ | Child's SUD rate | Ratio | Consumption equivalent |
|---|---|---|---|
| 0.00 | 8.92% | 1.00 | 0.0% |
| 0.20 | 12.91% | 1.45 | 45.8% |
| 0.57 (calibrated) | 26.76% | 3.00 | 303.5% |
| 0.60 | 28.30% | 3.17 | 343.5% |
| *Panel B. Does the externality reverse the ranking?* | | | |
| Private cost of conditionality | | | 0.105% |
| Disorder cases prevented | | | 2.492 pp |
| Consumption equivalent to the child, one case | | | 204% |
| Externality gain from conditionality | | | 5.086% |
| Gain over loss | | | 48.4 |

Note: Panel A is at the baseline programme, $\pi_V = 0.27$, since the transmission ratio in the data comes from an economy with a rationed programme; Panel B is at universal cover, one child per household, no discounting across generations. Every consumption equivalent here is the proportional rise in consumption, at every date and in every state, worth as much as the difference in lifetime value, so it is a compensating variation and can exceed one hundred percent.

### 3.6.2 An externality on children reverses it

I model an externality that runs through income rather than through preferences. I have considered two alternatives, but both seem to be at odds with reality. Letting the child of a user inherit a stronger taste, a lower $b$, matches the transmission fact but has the wrong welfare sign: $b$ enters only through $\theta \log(1 + v/b)$, so a lower one raises the child's objective at every $v > 0$ and its lifetime value with it. So the specification says that a child who inherits a stronger taste for drugs is *made better off* by inheriting it, and that is the opposite of what these children report. Growing up with household substance abuse is one of the adverse childhood experiences in Felitti et al. (1998), and it predicts higher rates of depression, worse self reported health and lower life satisfaction in adulthood; the children of parents with a SUD also have lower educational attainment, which is the premise of the transmission literature $\delta_d$ is calibrated against (McGovern et al., 2023). A model in which the transmitted trait raises the child's own welfare cannot be describing any of that, whatever it does to the transmission rate. Letting it inherit a greater susceptibility, a higher $\lambda$, fails another way, since a child facing a riskier drug optimally uses *less* ex ante, which is at odds with the data: children of parents with a SUD are *more* likely to use, not less.

Income works on both counts, and it works through the channels explored in Section 3.4: lower permanent income is unambiguously bad to receive, and this model already puts users at the bottom, so a child born poorer is mechanically more likely to use, which is aligned with the data. It is also supported by the evidence, and it stands in for more than income alone: a parent who diverts spending and time to drugs invests less in the child's education and development, and the income penalty is the reduced form of all of that (Chetty et al., 2016; McGovern et al., 2023). The income gradient does the whole job, so the model transmits a SUD across generations *without transmitting any preference for drugs.*

A child's baseline income is drawn from the same lognormal as everyone else, and the entire link is the penalty

$$y_{\text{child}} \;=\; y_{\text{base}}\,(1-\delta_d)^{\mathbf{1}\{\text{parent acquired the SUD}\}}\,(1-\delta_h)^{\mathbf{1}\{\text{parent was homeless}\}}, \tag{11}$$

which the parent does not internalise. That is what makes it an externality rather than something already priced into $V_1$.[9]

The disorder penalty $\delta_d$ is calibrated rather than set: it is chosen so that the child of a parent who acquired the SUD is 3 times as likely to acquire one, the middle of the epidemiological range of two to four in McGovern et al. (2023) and the alcohol literature. That takes $\delta_d = 0.57$, shown in Table 4, and it is worth saying that this is a larger penalty than anything the earnings literature measures directly: Nilsson (2017) finds 24 percent and Chetty et al. (2016) 31. The reason is that the model is being asked to reproduce an epidemiological ratio rather than an earnings coefficient, and in this model the only channel from parent to

[9]In the quantitative model the altruism of parents towards their children is calibrated rather than set to zero, so that the channel where a parent does take some account of what it does to its child is present and only the remainder is external.

child is income, so income has to carry whatever the childhood does. If part of the transmission runs through something else, through the home rather than the budget, then $\delta_d$ is standing in for more than earnings and should be read as the reduced form of the whole childhood rather than as an estimate of a wage effect.

The homelessness penalty $\delta_h$ is calibrated the same way, on the transmission of its own condition, so the two penalties are matched to the same kind of moment rather than one to a ratio and the other to a coefficient. Bassuk et al. (1997) is a case control study of 220 homeless mothers and 216 low income housed mothers in Worcester, Massachusetts: 14.2 against 8.3 percent had a primary female caretaker who had been homeless, and 10.3 against 4.4 percent a male one, which are odds ratios of 1.80 and 2.41. Both arms are poor, so the comparison isolates transmission from the poverty that accompanies it and needs no population base rate. $\delta_h$ is set so the child of a homeless parent is 2.10 times as likely to be homeless itself, the midpoint of the two, which takes $\delta_h = 0.33$.

Chetty et al. (2016) is then a check the calibration never used. They find that children moved to a lower poverty neighbourhood before age thirteen earn 31 percent more as adults, and the penalty that matches Bassuk implies 33 percent. An earnings estimate and a transmission ratio, from unrelated data and different countries, pin the same parameter to within two points. The identification runs the less comfortable way, since Bassuk is observational and single city where Chetty is quasi experimental, and that is worth saying: the target is the weaker design and the check the stronger. What makes the ordering right anyway is that $\delta_h$ exists to price the transmission of homelessness, and Bassuk measures that, where Chetty measures what a poorer neighbourhood does to earnings and only stands in for it.

The two penalties also do different work: both policies drive homelessness to zero at universal cover, so $\delta_h$ prices the programme against no programme while $\delta_d$ decides Treatment First against Housing First.

Panel B puts the two sides together and they are not close. The private cost of conditionality, in the first row, is 0.105 percent of consumption, small because only 4.8 percent of households are affected. Against that, conditionality prevents 2.492 percentage points of SUD cases in the second row, and preventing one of them is worth 204.1 percent of consumption to one child, shown in the third row. The gain in the fourth row is the product of those two, since one prevented case spares one child: 2.492 percentage points times 204.1 percent gives 5.086 percent of consumption, which is 48.4 times the private loss of the first row. That ratio exceeds one for every $\delta_d$ in Panel A except $\delta_d = 0$, where by construction there is no externality to price; at the smallest penalty shown it is still 8 times the private loss. Housing First is better for every household taken one at a time, and worse once one counts the children of the households it fails to deter.

All of this is a crude illustration: the model is two periods long and nobody who faces the sobriety condition fails to meet it, so the impacts of policies can be unreasonably large. It shows that an externality can reverse a welfare ranking that is otherwise airtight, which is all it is for. Which way the comparison actually goes is a question about magnitudes, and that is what the quantitative model of the next section is for.

The quantitative model carries two further channels absent here, and they are different in kind as well as in sign, which is another substantive reason the welfare ranking is a question of magnitudes. The first is not an externality at all but the distortions the programme creates and the distortionary taxes required to pay for it; the second is an externality of the same type as the one on children.

- **Programmes distort the decisions of recipients and are paid for out of distortionary taxes.** A programme lowers the labour supply of the households it does reach, because eligibility is means tested, and the voucher is clawed back at a rate of $\kappa_r$, so a recipient who works more loses eligibility or some of the transfer. This, combined with the pure financing of the programmes, means that other households have to pay more taxes, which distorts their labour supply.
- **Homelessness is a disamenity in itself.** Households dislike homelessness around them and not only their own, so the aggregate rate is an argument of everyone's utility. This one is an externality, and it survives any way of paying for the programme. It runs in favour of whichever policy houses more, which is the opposite direction from the tax.

# 4 The quantitative model

This section describes the model the paper's numbers come from. Where the stylized model already carries something unchanged it is named rather than motivated a second time: the Stone-Geary form of the drug term, the logic that separates the two policies, and the transmission externality all appear below. Notation follows Section 3 wherever the two models share an object: $\gamma$ and $\theta$ are the housing and drug weights in both, $b$ is the same Stone-Geary endowment, $\aleph$ the same flow value of being alive, $\lambda$ the same SUD hazard, $\psi$ and $\bar{y}$ the same replacement rate and earnings base cap, $r$ the same risk free rate, and $\beta$ is the discount factor and nothing else.

## 4.1 Demographics, the state and the choice set

Households enter as adults, live at most $J$ annual periods and retire at $J_R$, and $j$ indexes age within that span. Survival from $j$ to $j+1$ is $\phi(j, h', d, m, \tau)$ in (24), and it depends on age, on the housing *chosen* this period, on the SUD and on SMI, and on whether the household is in treatment, since treatment means it is not using, and thus is less likely to die. Newborns replace the dead exactly, so the mass of households is constant. The individual state is

$$s = (a,\, y,\, e,\, d,\, m,\, \eta) \in \mathcal{A} \times \mathcal{Y} \times \{1,2,3\} \times \{0,1\}^3, \tag{12}$$

where $a$ denotes assets, $y$ permanent productivity, $e$ employment status, $d$ the SUD, $m$ SMI and $\eta$ receipt of a voucher.

Each period the household chooses

$$\big(n,\, h',\, v,\, \tau,\, a'\big) \in [0,1] \times \{h_1, h_2, h_3, h_4\} \times \mathbb{R}_+ \times \{0,1\} \times \mathcal{A}, \tag{13}$$

where $n$ denotes hours, $h'$ the housing unit chosen, of which $h_1$ is homelessness, $v$ drug use, $\tau$ SUD treatment take up, available only when $d = 1$ and requiring $v = 0$ in the period it is taken, and $a'$ next period assets.

## 4.2 Preferences

Period utility is

$$u\big(c, h', v, n, \tau; d, m\big) \;=\; \frac{\left[\, c^{1-\gamma-\theta}\,(h')^{\gamma}\,(v+b)^{\theta}\,\right]^{1-\sigma}}{1-\sigma} \;+\; \omega_\ell \frac{(1-n)^{1-\sigma_\ell}}{1-\sigma_\ell} \;+\; \aleph \;-\; \tau\bar{\tau}^{u} \;-\; \kappa_x H, \tag{14}$$

Four of these are the stylized model's and carry the same meaning here as in Section 3.2: $\gamma$ the housing weight, $\theta$ the drug weight, so that $1-\gamma-\theta$ is the weight on consumption and the three sum to one; $b$ the Stone-Geary endowment; and $\aleph$ the flow value of being alive. Equation (1) is the logarithmic case of this bundle with hours fixed.

The rest are new, and there are three of them. Leisure makes hours a margin, at weight $\omega_\ell$ and curvature $\sigma_\ell$ over the $1-n$ hours not worked. SUD treatment costs utility, $\bar{\tau}^u$ in each period the household takes it up, $\tau = 1$. And homelessness carries a disamenity, $\kappa_x$ per unit of the aggregate homelessness rate $H$, borne by every household and priced on the headcount alone, which is what makes homelessness an externality here rather than only a private misfortune; Section 5.1 takes it up. Outside the bundle, $\sigma$ is the coefficient of relative risk aversion.

## 4.3 Income, taxes and transfers

Gross income depends on age, employment, productivity and health,

$$y^g(j,y,e,d,m,n) = \begin{cases} 0.40\,y & j \geq J_R \quad \text{(social security)} \\ n\,y\,\varepsilon(j)\,\rho_d^{\mathbf{1}\{d=1\}}\rho_m^{\mathbf{1}\{m=1\}} & j < J_R,\ e=1 \quad \text{(employed)} \\ \psi\ \min\{y\,\varepsilon(j),\ \bar{y}\} & j < J_R,\ e=2 \quad \text{(first year of a spell)} \\ \zeta_{DI}\,y\,\varepsilon(j) & j < J_R,\ e=3,\ m=1 \quad \text{(disability)} \\ 0 & \text{otherwise,} \end{cases} \tag{15}$$

taking the cases in the order they appear. Retirement income replaces 40 percent of the productivity state with no health penalty, since social security is not means tested on health. For the employed, $\varepsilon(j)$ is the deterministic age profile of efficiency and $\rho_d$ and $\rho_m$ are productivity penalties applied *conditional on working*. The three employment states are work, $e = 1$, the first period of a spell out of work, $e = 2$, and non employment thereafter, $e = 3$, so the last two cases are both unemployment and the disability case is the income of a household that is out of work and carries SMI. Unemployment insurance (UI) is paid in the first period of a spell only, at the replacement rate $\psi$ on an earnings base capped at average income $\bar{y}$, both as in equation (4). $\zeta_{DI}$ is the rate at which disability insurance (DI) replaces the income of a non employed household with SMI.

The two penalties, $\rho_d$ and $\rho_m$, are in the model to generate a large share of homeless households being employed, consistent with the data. Employed homelessness cannot be produced by unemployment alone: something has to leave a household poor while it is still in a job. Both are calibrated internally: $\rho_d$ on the share of the homeless who are employed, and $\rho_m$ on the earnings of workers with SMI relative to average.

Labour income is taxed with the Heathcote et al. (2017) function and capital income at a flat rate $\tau_k$,

$$y^n = \tau_\ell\,(y^g)^{\xi} \ +\ (1-\tau_k)\,r\,a, \tag{16}$$

with $y^n$ denoting after tax income, where $\xi$ governs progressivity and $\tau_\ell$ the level. Households can receive two transfers, both means tested on gross income as the programmes themselves are. A food stamp floor tops means tested income up to $\underline{c}\,\bar{y}$, with $\underline{c}$ the floor expressed as a share of average income, and the voucher covers rent above a tenant contribution of $\kappa_r$ of gross earnings, the same $\kappa_r$ as in Section 3.2.3,

$$F = \max\big\{0,\ \underline{c}\,\bar{y} - (y^g + ra)\big\}, \qquad Vch = \mathbf{1}\{\eta=1\}\,\mathbf{1}\{h' \neq h_1\}\,\max\big\{0,\ p_r\,h' - \kappa_r\,y^g\big\}. \tag{17}$$

A homeless voucher holder does not lose the voucher: retention depends on income eligibility, not on housing, so the entitlement survives and simply pays nothing until the household rents again.

## 4.4 The budget constraint

$$c + p_v v + a' \ =\ y^n + a \ -\ p_r\,h'\,\mathbf{1}\{h' \neq h_1\} \ +\ Vch + F, \qquad c > 0,\ v \geq 0,\ a' \geq 0. \tag{18}$$

The drug price is $p_v$ and market rent is $p_r$, both as in Section 3.2.3, and $a$ is assets carried in. Note what is absent: treatment, which costs utility, and bequests, which go to the government. How $c$, $v$ and $n$ are chosen out of it is with the rest of the optimisation, in Section 4.11.

## 4.5 Housing, and the homelessness loop

Housing is a four point grid. $h_1$ is literal homelessness, sleeping rough or in an emergency shelter, which is the HUD point-in-time definition and excludes doubling up; it carries a small service flow and no rent, a fixed fraction of the smallest rental rather than a level of its own. The three rental sizes are pinned at both ends and spaced by a curvature between them,

$$h_2 = \underline{h}, \qquad h_3 = \underline{h} + \Big(\tfrac{1}{2}\big(\bar{h} - \underline{h}\big)^{1/\varsigma}\Big)^{\varsigma}, \qquad h_4 = \bar{h}, \tag{19}$$

so three parameters describe the grid: $\underline{h}$ the smallest rental size, $\bar{h}$ the largest, and $\varsigma$ the curvature that places the rung between them. At $\varsigma = 1$ the ladder is uniform and at $\varsigma > 1$ the middle rung sits closer to the bottom, which is what a housing size distribution with most households in the largest category requires.

Homelessness then enters three separate equations: survival (24), the employment transitions (26) and the SUD recovery hazard (21), through $\kappa_{\phi H}(j)$ and $\kappa_{ueH}$ in Section 4.8 and $\mu_H$ in Section 4.6. It raises mortality, worsens both employment margins and lowers the chance of recovering from the SUD. Mental illness is the one thing it does not reach. The SMI chain runs on its own, and that is a finding rather than an omission: the trial that housed people with serious mental illness looked for an effect on their symptoms and did not find one, while finding large effects on their housing. Section 4.7 gives it.

## 4.6 Drug use, the SUD, recovery and treatment

A household either abstains or uses, with $v$ chosen continuously. A clean household that uses acquires the SUD with the hazard of equation (2),

$$\Pr(d' = 1 \mid d = 0, v) = \pi_d(v) = 1 - e^{-\lambda v}, \tag{20}$$

In the baseline a clean household may initiate only up to an age $J_E$, the last age at which use can begin.[10] A household with the SUD recovers with probability

$$\pi_r(h', m, \tau) = \min\left\{1,\ \bar{\pi}_r\, \mu_H^{\mathbf{1}\{h'=h_1\}}\, \mu_m^{\mathbf{1}\{m=1\}}\, g^{\tau}\right\}, \tag{21}$$

with $\bar{\pi}_r$ the baseline annual chance of recovering while housed, clean of SMI and untreated, and three multipliers on it: $\mu_H < 1$, so recovery is harder on the street, $\mu_m < 1$, harder with SMI, and $g > 1$ the treatment effect.

$\mu_H$ is how housing acts on drug use here, and it is worth being exact about what that claim is and is not. Housing does not change what a household wants: flow utility is the same function of $(c, h, v)$ housed or not. What housing changes is the chance of getting out of the disorder once it is carried, recovery on the street being $\mu_H$ times recovery while housed. Milby et al. (2005) randomised cocaine dependent homeless adults into arms differing in housing and measured weeks of abstinence, so the ratio between arms identifies $\mu_H$ in our calibration.

Treatment $\tau$ is a *choice*, not a mechanical hazard, taken at the utility cost $\bar{\tau}^u$ in (14). It also costs the drug: a household in treatment must abstain in that period, $v = 0$ whenever $\tau = 1$, which is imposed as a constraint on (30) rather than priced. On the other hand, because it is not using, its mortality rate goes back to the baseline for that period.

$g$ is the parameter through which its entire effect runs, so treatment has to be something a household decides to undertake and can decline.

## 4.7 Serious mental illness

$m$ is an exogenous two state Markov chain representing SMI, whose prevalence $p_m(j)$ averages about 5.5 percent of adults, defined by substantial functional impairment, rather than any mental illness, which is roughly a quarter of adults and includes mild anxiety: a state that halves productivity cannot sensibly cover a quarter of the population.

The onset rate is a function of age. Writing $p_m(j)$ for prevalence at age $j$ and $\lambda_m$ for the annual chance of leaving the state,

$$\text{onset}_m(j) = \frac{p_m(j{+}1) - p_m(j)(1 - \lambda_m)}{1 - p_m(j)}, \tag{22}$$

which is the accounting identity that reproduces the observed age profile exactly, backed out of the age profile of prevalence in NSDUH 2023 and the recovery hazard. The transitions are then

$$\Pr(m' = 1 \mid m = 0, d) = \text{onset}_m(j) \cdot \iota^{\mathbf{1}\{d=1\}}, \qquad \Pr(m' = 0 \mid m = 1) = \lambda_m, \tag{23}$$

[10] Without it there is little consequence at old ages to using: too little life remains for the disorder to damage, so the model produces too much of it.

where $\iota > 1$ is the multiplier on onset for a household carrying the SUD, which makes a SUD raise the hazard of SMI. That last channel runs in the direction the evidence supports: substance induced psychosis converts to schizophrenia or bipolar disorder at a pooled rate near a quarter (Murrie et al., 2020), and in the Danish registers any SUD predicts conversion to schizophrenia with a hazard ratio of 1.34, opioid use disorder 2.74 (Nielsen et al., 2017).

SMI then affects five objects: productivity conditional on working through $\rho_m$ in (15), both employment margins through $\kappa_{uem}$ in (26), mortality through $\kappa_{\phi m}$ in (24), the SUD recovery hazard through $\mu_m$ in (21), and disability income, for which eligibility is being out of work with SMI and the replacement rate is $\zeta_{DI}$. It does not change the utility function.

The mental health chain does *not* depend on housing, and that is a finding in the data rather than an omission. In the At Home / Chez Soi randomised trial, 1,198 homeless adults with SMI across four cities, assignment to housing with intensive case management produced no detectable effect on symptom severity at any wave, while producing large effects on housing stability, its primary outcome. Making the channel housing dependent would mean adding a mechanism the best available trial looked for and did not find.

## 4.8 What the SUD, SMI, and homelessness cost

**Death risk.** Survival is

$$\phi(j, h', d, m, \tau) = 1 - \min\Big\{1,\ (1 - \bar{\phi}(j))\, \kappa_{\phi H}(j)^{\mathbf{1}\{h'=h_1\}}\, \kappa_{\phi d}(j)^{\mathbf{1}\{d=1\}\mathbf{1}\{\tau=0\}}\, \kappa_{\phi m}^{\mathbf{1}\{m=1\}}\Big\}, \tag{24}$$

with $\bar{\phi}(j)$ the baseline survival age profile and $\kappa_{\phi H}(j)$ and $\kappa_{\phi d}(j)$ two age varying multipliers, and $\kappa_{\phi m}$ one that is not, raising the death hazard for homelessness, the SUD and SMI respectively. The SUD multiplier is switched off in a period the household is in treatment, $\tau = 1$, which is why the survival function takes $\tau$ as an argument at all. The excess mortality of the disorder is overwhelmingly a mortality of use, poisoning above all, and treatment forbids use in the period it is taken, so a household that is not using does not carry it. I make this assumption because that is what the cause specific evidence shows. The meta analyses of mortality among people using extramedical opioids report overdose as the largest single contributor to the excess (Larney et al., 2020; Degenhardt et al., 2011), and in CDC WONDER drug poisoning alone accounts for about a fifth of all deaths at ages 25 to 44, which is the moment Section 5.3 reports the model against. $\bar{\phi}(j)$ is the SSA period life table for ages 26 to 85, with the sexes combined by survivor weight, $q = [l^M q^M + l^F q^F]/[l^M + l^F]$.

The homelessness multiplier cannot be read off the data directly, and the reason is worth spelling out. Baggett et al. report a *standardised mortality ratio* (SMR) for homeless adults: deaths among the homeless at a given age divided by deaths in the general population at that age, which they publish by age band and which we write $RR(j)$. It cannot be used as $\kappa_{\phi H}(j)$ because the people it is measured on are not only homeless. A share $s = 0.217$ of them also carry a SUD (Gutwinski et al., 2021), which raises mortality by itself, so the published ratio is a mixture of two effects and attributing all of it to housing would overstate what housing does.

Three things are observed and two are solved for. Observed: $RR(j)$, Baggett's ratio; $\sigma_d(j)$, the shape of the SUD's mortality effect over age; and $s$, the SUD share of the homeless. Solved for: $\kappa_{\phi d}(j)$ and $\kappa_{\phi H}(j)$, one age profile from each equation. Picking the single number $\bar{\kappa}_{\phi d}$ settles both: the first equation turns it into $\kappa_{\phi d}(j)$ at every age, and the second then reads $\kappa_{\phi H}(j)$ off Baggett's ratio at every age.

Undoing the mixture needs the SUD's own multiplier at each age. That is what $\sigma_d(j)$ supplies: the age profile of drug poisoning deaths from CDC WONDER 2023, normalised to one at the age where it peaks, so $\sigma_d(j)$ is the share of the SUD's peak mortality effect that is present at age $j$. Writing $\bar{\kappa}_{\phi d}$ for the level of the SUD multiplier,

$$\kappa_{\phi d}(j) = 1 + (\bar{\kappa}_{\phi d} - 1)\, \sigma_d(j), \qquad \kappa_{\phi H}(j) = \max\Big\{1,\ \frac{RR(j)}{(1-s) + s\, \kappa_{\phi d}(j)}\Big\}. \tag{25}$$

The denominator is the average mortality multiplier in Baggett's population: a share $1 - s$ of it carries homelessness alone and a share $s$ carries the SUD as well. Dividing the observed ratio by it leaves the homelessness effect on its own.

$\bar{\kappa}_{\phi d}$ is assigned from the literature on all cause standardised mortality ratios for opioid use disorder, which report values between 6 and 10 (Degenhardt et al., 2011; Hser et al., 2017; Larney et al., 2020), and the calibration section reports the value taken. The drug price participation elasticity is not targeted; it is reported as a validation moment and compared with the range estimated in the literature, $-0.3$ to $-1.1$ (Saffer and Chaloupka, 1999; Grossman and Chaloupka, 1998; Olmstead et al., 2015).

$\kappa_{\phi m}$ carries no age argument because the data give it basically none. Olfson et al. (2015) give all cause standardised mortality ratios for schizophrenia of 3.6, 3.4 and 4.0 across ages 20 to 34, 35 to 54 and 55 to 64. Flat, if anything rising.

**Unemployment risk.** Job finding and separation are

$$\pi^{ue} = \min\left\{1,\ \bar{\pi}^{ue}\,\kappa_{ueH}^{\mathbf{1}\{h'=h_1\}}\kappa_{ued}^{\mathbf{1}\{d=1\}}\kappa_{uem}^{\mathbf{1}\{m=1\}}\right\}, \qquad \pi^{eu} = \min\left\{1,\ \bar{\pi}^{eu}\,\kappa_{ueH}^{-\mathbf{1}\{h'=h_1\}}\kappa_{ued}^{-\mathbf{1}\{d=1\}}\kappa_{uem}^{-\mathbf{1}\{m=1\}}\right\}, \tag{26}$$

with $\bar{\pi}^{ue}$ and $\bar{\pi}^{eu}$ the baseline job finding and separation rates and $\kappa_{ueH}$, $\kappa_{ued}$ and $\kappa_{uem}$ the multipliers for homelessness, the SUD and SMI, so each multiplier below one makes work simultaneously harder to find and easier to lose.

**The child's income risk.** A parent with a SUD, or one who ends up on the street, lowers the productivity ($y_{\text{child}}$) its child is born with, through $\delta_d$ and $\delta_h$ respectively, as shown in (39) in Section 4.14. That is a risk to the household's own objective only to the extent that $\omega$, the weight the parent places on its child's lifetime utility in (30), is positive. Since $\omega$ is small (see Section 5 for details), the household weighs a fraction $\omega$ of this risk and $1-\omega$ of it falls outside the problem entirely. It is therefore the one risk here that is principally an externality rather than a pure personal cost, which is what makes it a candidate for policy in a way that mortality and unemployment risk, both fully internalised, are not.

## 4.9 The voucher, and the two policies

A household without a voucher draws one with probability $\pi_V$, the same lottery probability as in Section 3.2.3, if it is income eligible, and a household with one keeps it while eligible, so unlike Section 3.2.3 the voucher is persistent rather than redrawn each period. Eligibility is

$$y^g + ra \;<\; \kappa_V\, y^{\text{med}}. \tag{27}$$

with $\kappa_V$ the eligibility cutoff expressed as a share of median gross income $y^{\text{med}}$. Under Housing First that is the whole rule. Under Treatment First the voucher is additionally lost with probability $p_{\text{det}}$ by a household that uses drugs or has a SUD, and with probability $p_{\text{fp}}$ by one that did not,

$$\Pr(\eta'=1) = \Pr(\eta'=1)^{HF} \times \begin{cases} 1-p_{\text{fp}} & \text{if } (v=0 \text{ and } d=0) \text{ or } \tau=1, \\ 1-p_{\text{det}} & \text{otherwise.} \end{cases} \tag{28}$$

There are four cases and the rule covers them. A clean household that does not use meets the condition and faces only the false positive rate $p_{\text{fp}}$. A clean household that uses fails it and faces $p_{\text{det}}$. A household carrying the disorder that does not take treatment fails it and faces $p_{\text{det}}$ as well. And a household carrying the disorder that does take treatment is abstaining, $v=0$ by the restriction in Section 4.6, so it meets the condition and faces $p_{\text{fp}}$. There is no fifth case: treatment is available only when $d=1$, and it forbids use, so no household is ever both in treatment and using.

This is the one place where the quantitative model changes the *logic* of Section 3.5 rather than its setting. There, conditionality is perfectly enforced and perfectly met, so Treatment First deters everyone and nobody is thrown off the rolls. Here detection is imperfect and so are households: some fail the condition because they cannot get clean, and some fail it because the test is wrong. Those two groups are the empirical case for Housing First, neither exists in the stylized model, and that is why its welfare numbers were illustration.

## 4.10 The shocks, collected

Conditional on the state and the choices, next period's state is drawn from five independent transitions. Collecting them in one place:

**Productivity $y$.** Fixed over the life cycle, so $y' = y$. I do this for two reasons. First, it is what keeps the child's penalty (39) a shift along a single scalar rather than a shift across the nodes of an age varying discretisation, which significantly reduces the computation time to manageable levels. Second, it still captures the key mechanisms, because unemployment and the penalty $\rho_d$ are the income risks that link the SUD to housing, and those are retained in full.

**Employment $e$.** A household that loses their job receives insurance with certainty for one year. With $\pi^{ue}$ and $\pi^{eu}$ from (26), it gives the following transitions:

$$e = 1 \;\longrightarrow\; \begin{cases} e' = 1 & \text{w.p. } 1 - \pi^{eu} \\ e' = 2 & \text{w.p. } \pi^{eu} \end{cases} \qquad e \in \{2, 3\} \;\longrightarrow\; \begin{cases} e' = 1 & \text{w.p. } \pi^{ue} \\ e' = 3 & \text{w.p. } 1 - \pi^{ue}, \end{cases} \tag{29}$$

State 2 is reachable only from employment and never from itself, so a household is in it for the first year out of work and in state 3, which pays nothing, from the second year on. One year is if anything generous: most states pay for twenty six weeks.

**The SUD $d$.** By (20) if $d = 0$ and (21) if $d = 1$, conditioning on the chosen $v$, $h'$ and $\tau$.

**Mental illness $m$.** By (23), conditioning on $d$.

**The voucher $\eta$.** By (27) and (28), conditioning on next period's eligibility and, under Treatment First, on the chosen $v$, the status $d$ and the treatment choice $\tau$.

Write $\Gamma\big(s' \mid s,\, n, h', v, \tau\big)$ for the product of these five, which is the object the expectation in (30) is taken against.

## 4.11 The household's dynamic problem

Putting it together, for $j = 1, \ldots, J$ and with $V(J{+}1, \cdot)$ given below,

$$\begin{aligned} V(j, a, y, e, d, m, \eta) = \max_{\{n,\, h',\, v,\, \tau,\, a'\}} \Bigg\{ & \frac{\big[c^{1-\gamma-\theta}(h')^{\gamma}(v+b)^{\theta}\big]^{1-\sigma}}{1-\sigma} + \omega_{\ell}\frac{(1-n)^{1-\sigma_{\ell}}}{1-\sigma_{\ell}} + \aleph - \tau\bar{\tau}^{u} - \kappa_x H \\ & + \; \beta\,\phi(j, h', d, m, \tau)\int_{\mathcal{S}} V(j{+}1)(s')\,\Gamma\big(\mathrm{d}s' \mid s, n, h', v, \tau\big) \\ & + \; \beta\,\big[1 - \phi(j, h', d, m, \tau)\big]\,\omega\,\Big[\mathbb{E}\,V\big(1, 0, y_{\text{child}}, e', 0, 0, 0\big) \\ & \qquad - \; \mathbb{E}\,V\big(1, 0, y_{\text{base}}, e', 0, 0, 0\big)\Big] \Bigg\} \end{aligned} \tag{30}$$

subject to the budget (18), the income definitions (15) to (17), the choice set of dwellings (13), $\tau = 0$ whenever $d = 0$, $n = 0$ whenever $j \geq J_R$ or $e \neq 1$, $v = 0$ whenever $\tau = 1$, $a' \geq 0$, $c \geq F$ so that consumption never falls below the food stamp floor, and $V(J{+}1, \cdot) = 0$ with $\phi(J) = 0$, so that at the last age the child is the whole continuation. The integral runs over next period's state $s' = (a', y, e', d', m', \eta')$; the second line of (30) is the household's own continuation, and the third is the child's expected lifetime utility, entered at age one with zero assets and with its own employment status $e'$ drawn as any newborn's is.

A household that does not survive from $j$ to $j + 1$ leaves a child, so the continuation is the survival weighted value of its own next period plus the death weighted value of what the household's own state costs

that child, each discounted, with $y_{\text{child}}$ as defined in (39). What that buys is a second route by which the household's own decisions, to use and to go unhoused, reach its child. $\phi(j,\cdot)$ itself depends on the dwelling chosen, so the housing decision moves the weight on the two terms directly. The drug does not move it within the period, because $\phi$ takes the disorder as a state and $v$ only raises the chance of carrying it *next* period. The drug reaches the child through the continuation instead, which is where the first order condition below picks it up.

The third line is a *difference* and not a level, and the distinction is not cosmetic. Written as a level, $\omega$ times the child's utility, the term is worth more to a household the likelier it is to die, because it is the death branch that carries it. A SUD raises the death hazard several fold, so a level would pay most to exactly the households the paper is about, and it would do so for reasons that have nothing to do with the child.

Subtracting the value of a child born to a clean, housed parent removes that. The term is zero for a household in the reference state and negative otherwise, so what enters the decision is only the loss the parent's own state imposes on its child, weighted by the chance that child is born at all.

At $\omega = 0$ the damage a household does to its child is entirely external; at $\omega > 0$ the household internalises a share $\omega$ of it, and only $1-\omega$ remains external. Because the term is a difference, $\omega$ is the weight on that damage and nothing else. Since the child's productivity depends on whether the parent carried a SUD or was homeless, and both depend on $v$ and $h'$, the child term in (30) is what puts the child inside the parent's drug and housing decision.

What remains external is more than $1-\omega$ once one looks past the child. A child born with the penalty is likelier to develop a SUD and SMI, and be unhoused, when its own death draw arrives, so it is likelier to pass the penalty on, and the parent's decision could ripple through its entire lineage. Equation (30) does price that chain, but geometrically: the child's value $V(1,\cdot)$ on the third line contains its own $\omega$ weighted child term, so generation $k$ enters the parent's problem at $\omega^k$, times the discounting and the death probabilities along the way, and $1-\omega^k$ of it stays outside. The penalty itself does not compound, since every newborn draws afresh from $\Omega_y$ in (39) and takes at most one $\delta_d$ and one $\delta_h$; what compounds is the chance of taking it at all. So the object here is not a transfer from one parent to one child but a hazard running down a line of descendants. To save space, the rest of the paper does not restate that whole line every time the externality comes up, and speaks of the cost to the child instead. What is meant throughout is the lineage.

**How the choices separate, and where they do not.** Taking hours, the dwelling and next period assets as given, what is left to divide between consumption and the drug is

$$x \;=\; y^n + a \;-\; p_r\, h'\, \mathbf{1}\{h' \neq h_1\} \;+\; Vch + F \;-\; a', \tag{31}$$

so the budget (18) reads $c + p_v v = x$ and the whole of the remaining problem is one number: how to split $x$ between $c$ and $v$, since either one determines the other.

Were the drug an ordinary good, one whose only consequence is the flow utility it delivers this period, that split would follow from flow utility alone. It does not, because $v$ enters the transition as well as the utility: by (20) another unit of the drug also buys a higher chance of waking up next period with the disorder. So there are three terms in the first order condition for $v$, not two,

$$\underbrace{\frac{\theta}{b+v}\,\Lambda}_{\text{value of another unit}} \;=\; \underbrace{\frac{p_v\,(1-\gamma-\theta)}{c}\,\Lambda}_{\text{the consumption it displaces}} \;+\; \underbrace{\lambda e^{-\lambda v}\Big[\,\mathbb{E}V(\cdot\,|\,d'=0) - \mathbb{E}V(\cdot\,|\,d'=1)\,\Big]}_{\text{the risk it adds}}, \tag{32}$$

where $\Lambda = \left[c^{1-\gamma-\theta}(h')^{\gamma}(v+b)^{\theta}\right]^{1-\sigma}$ is the Cobb-Douglas bundle of (14). It multiplies the first two terms and not the third, because the third is a difference of continuation values and is therefore already in utils. Were it absent, $\Lambda$ would divide out of the condition entirely and the split would reduce to the usual Cobb-Douglas one; it is the third term that stops it cancelling, and that is why the marginal utility level matters here and not in an ordinary two good problem. That third term is the marginal hazard $\pi_d'(v) = \lambda e^{-\lambda v}$ times what the disorder costs, the gap between the two continuation values. That gap is where the externality on the child enters the drug decision, and it is why there is no fourth term. Each continuation value carries its own child term, at $d'=0$ and at $d'=1$, so the difference between them already contains the $\omega$ share of what the disorder costs the child alongside what it costs the household itself. A fourth term would be needed only

if using raised the chance the child's term is realised within the period, and it does not: $\phi$ depends on the disorder as a state, not on how much is bought today. This is Section 3's mechanism inside the quantitative model: because $\pi_d$ rises in $v$, how much to use is not decided on the budget alone.

There are two cases in which the third term is zero, and in both the household faces no such trade off: one that abstains, where $v = 0$, and one that already carries the disorder, whose recovery hazard (21) does not depend on how much it uses. For both the condition is the Cobb-Douglas split in closed form,

$$c = \max\Big\{F,\ \frac{1-\gamma-\theta}{1-\gamma}\big(x + p_v b\big)\Big\}, \qquad v = \max\Big\{0,\ \frac{x-c}{p_v}\Big\}, \tag{33}$$

whose corner is the abstention decision: a household buys nothing when the Stone-Geary endowment already exceeds what it wants, which is how a large $b$ deters use and is the threshold of Section 3.2.4. The outer maximum on $c$ is the food stamp floor. Where it binds the Cobb-Douglas split no longer describes the allocation: consumption is held at $F$ and the drug takes what the budget leaves, so the floor changes behaviour and not only welfare. Where the third term is present the answer lies between (33) and abstention, since risk can only push towards less, so it is found by a one dimensional search on that interval.

Hours then solve

$$u_c(c)\,\frac{\partial x}{\partial n} \;=\; \omega_\ell (1-n)^{-\sigma_\ell}, \tag{34}$$

and the clawbacks in (17), at a rate of one for food stamps and $\kappa_r$ for the voucher, make $\partial x/\partial n$ jump *up* at the hours where each transfer runs out, because the clawback stops. That is the welfare trap, and it is the channel through which conditionality changes labour supply rather than only consumption.

## 4.12 The government

The government pays social security, unemployment insurance, disability, food stamps and vouchers, and supplies a public good $\bar{G}$ that does not enter private consumption and is held fixed across simulations, so that policies are compared at a constant level of it. It finances this with the two taxes in (16) and the estates of the deceased, with the level $\tau_\ell$ clearing the budget in every simulation,

$$\underbrace{\textstyle\sum y^g - \tau_\ell \sum (y^g)^\xi + \tau_k \sum ra + \text{estates}}_{\text{revenue}} \;=\; \underbrace{SS + UI + DI + F + Vch + \bar{G}}_{\text{outlay}}. \tag{35}$$

In the stylized model the two policies happen to cost the same and the comparison is unaffected. Here a policy that houses more households spends more, $\tau_\ell$ rises to pay for it, and the higher tax lowers the after tax return to an extra hour: it enters (34) through $\partial x/\partial n$, which is net of the tax schedule in (16), so hours fall and saving with them, for everyone, including the households the programme never reaches, and it runs against Housing First.

## 4.13 The drug market

Drug production has cost $C(S_v) = \frac{\nu_1}{\nu_2} S_v^{\nu_2}$, where $S_v$ is the aggregate quantity. Supply satisfies

$$p_v = \nu_1 S_v^{\nu_2 - 1}, \qquad S_v = \int v(s)\,\mathrm{d}\Phi(s), \tag{36}$$

with price elasticity of supply $1/(\nu_2 - 1)$ and $\nu_1$ normalised so that the quantity supplied equals the quantity demanded at the benchmark price, which is normalised to one. The supply elasticity, its range and the evidence behind them are set in Section 5.

## 4.14 The law of motion and newborns

Given the policy functions, the distribution over states evolves by pushing households through their choices and then through the shocks,

$$\Phi_{j+1}(\mathcal{S}') \;=\; \phi(j,\cdot) \int \Gamma\big(\mathcal{S}' \mid s,\, n_j(s), h'_j(s), v_j(s), \tau_j(s)\big)\,\mathrm{d}\Phi_j(s) \qquad \text{for every measurable } \mathcal{S}', \tag{37}$$

together with the newborn inflow

$$\Phi_1(\mathcal{S}') \;=\; \int \big[1-\phi(j,\cdot)\big]\,\mathcal{N}\big(\mathcal{S}' \mid s\big)\,\mathrm{d}\Phi_j(s) \;+\; \int \mathcal{N}\big(\mathcal{S}' \mid s\big)\,\mathrm{d}\Phi_J(s), \tag{38}$$

where $\mathcal{N}$ places a newborn according to (39), so that the mass of newborns equals the mass of the dead exactly and the two equations together conserve the population.

Here $\phi(j,\cdot)$ is evaluated at the household's own $(h'(j,s),d,m)$. The mass that does not survive, plus the mass at $j=J$, is replaced by newborns at $j=1$ with zero assets, clean, without SMI and without a voucher, and subject to the same unemployment shocks as anyone else, whose productivity is drawn from the unconditional distribution $\Omega_y$ of permanent productivity across households scaled by the transmission penalties $\delta_d$ for a child whose parent carried the SUD and $\delta_h$ for one whose parent was homeless,

$$y_{\text{child}} \;\sim\; \Omega_y \;\text{ scaled by }\; (1-\delta_d)^{\mathbf{1}\{\text{parent had the SUD}\}}\,(1-\delta_h)^{\mathbf{1}\{\text{parent was homeless}\}}. \tag{39}$$

The evidence behind both penalties is set out in Section 1 and placed against the literature in Section 2, and it is worth saying here only what the functional form takes from it. Three separate links are being compressed into two numbers: that a parent's disorder changes the childhood, that a parental shock transmits to the child's own adult earnings, and that housing is one of the environments doing the transmitting. Only the last is measured as an earnings effect of the right kind, which is why $\delta_h$ is set directly to the 31 percent of Chetty et al. (2016) while $\delta_d$ is set to the value that reproduces the disorder ratio of McGovern et al. (2023). Neither is a removal effect: whether foster placement itself helps or harms is contested, and $\delta_d$ is a reduced form for the whole bundle of that childhood rather than for any one intervention in it.

### 4.15 Competitive equilibrium

A competitive equilibrium is a collection of

- value functions $\big\{\{V_t(j,\cdot)\}_{j=1}^{J}\big\}_{t=0}^{\infty}$;
- policy functions $\big\{\{n_t(j,\cdot),\, h'_t(j,\cdot),\, v_t(j,\cdot),\, \tau_t(j,\cdot),\, a'_t(j,\cdot)\}_{j=1}^{J}\big\}_{t=0}^{\infty}$;
- distributions $\big\{\{\Phi_t(j,\cdot)\}_{j=1}^{J}\big\}_{t=0}^{\infty}$; and
- prices, policies and aggregates $\big\{\tau_{\ell t},\, p_{vt},\, \bar{y}_t,\, y_t^{\text{med}},\, H_t\big\}_{t=0}^{\infty}$,

such that, at every date $t$,

1. the policy functions solve (30) given its terminal condition, and $\{V_t(j,\cdot)\}_{j=1}^{J}$ are the associated values;
2. $\Phi_{t+1}(j{+}1,\cdot)$ follows from $\Phi_t(j,\cdot)$ by the law of motion (37) together with the newborn rule (39) and (38), and the mass of newborns equals the mass of the deceased;
3. the government budget (35) holds at $\tau_{\ell t}$;
4. the drug market clears at $p_{vt}$, so (36) holds with aggregate drug demand $S_{vt}=\int v_t(s)\,\mathrm{d}\Phi_t$;
5. the public good is supplied at its assigned level, $G_t=\bar{G}$;
6. $\bar{y}_t$, median gross income $y_t^{\text{med}}$ and the homelessness rate $H_t$ are the aggregates implied by $\Phi_t$.

A stationary equilibrium is the special case in which all of these are constant and $\Phi_{t+1}=\Phi_t$, which is what the calibration and the policy comparisons report. The sequence version is stated for completeness, and because the transition exercise in the results needs it.

## 5 Calibrating the quantitative model

The model has fifty three parameters, some of which vary in value by age. Twenty eight are calibrated externally, from data or from the literature, and twenty five are calibrated internally, inside the model. Four further parameters govern drug testing and detection; they are inert in the benchmark, where nothing is conditional, so they are set with the Treatment First experiment rather than here. They are split across

the two tables of this section: twenty nine are the main story of this paper and its policy analysis (Table 5, sixteen internally and thirteen externally calibrated), and twenty four are standard (Table 6, eight internally and sixteen externally).

The tables carry the parameter, its description, the moment and source used to fix its value, and the value itself. The text alongside each table adds what a table has no room for: the intuition for what each parameter does, and why its moment identifies it. Every estimate that goes into them is documented in full in the replication package: `data/DATA_APPENDIX.md` names the dataset behind each target, its year and universe, and the computation that turns it into the number reported here, so that any of them can be reproduced from the public microdata without reference to this text. The package also carries the datasets themselves, so nothing has to be downloaded to reproduce any of this: the extracts behind every target sit next to the code that reads them.

## 5.1 The twenty nine parameters that carry the story of this paper

Twenty four of the fifty three are common to most life cycle models, whereas the rest are specific to this paper, and all twenty nine of those carry its main story and policy analysis. They are set out in Table 5, with the parameter, its description, and the moment and source used to fix its value, which is given in the last column. They are discussed one at a time below because they are the parameters a reader may want to argue over, and each is taken in the order of the table.

The comparison between Housing First and Treatment First has a shape, and every entry below is read against it, so it is worth setting out this comparison once and in full before the entries begin.

Three things decide the ranking between Housing First and Treatment First. The **private value** is what a household itself gains from being housed and from not carrying a disorder: consumption, a dwelling rather than the street or shelter, and above all the years of life that homelessness and the disorder take away. The **externality** is what lands on somebody else: on the child, who inherits not the disorder or homelessness but a worse productivity draw associated with the parent's SUD and with the parent's homelessness, which $\delta_d$ and $\delta_h$ price separately. This means that the child is likelier in turn to be poor, to use, to acquire a disorder and mental illness as a consequence, to be unhoused and to die young. The externality also lands on everybody, through the disamenity of aggregate homelessness. The parent internalises only $\omega$ of the part that falls on its child, and $\omega$ is small, which is what leaves something for a policy to correct. And the **distortion** is what the programmes cost in behaviour, and it comes from two places. The programmes create distortions due to their design: assistance is means tested and the rent share is clawed back at $\kappa_r$, so a recipient who works more loses eligibility or part of the transfer, which requires even higher taxes on everyone else to fund even larger programmes. And assistance is paid for by distortionary taxes, which lower the after tax return to an hour for households neither policy ever reaches.

*Housing First is redistribution, removes externalities, but creates another.* Mechanically it removes two externalities: a child of a housed parent does not carry the penalty $\delta_h$ prices, and aggregate homelessness falls, which every household values through $\kappa_x$. However, it creates one at the same time, because assistance that arrives whatever a household does insures it against the consequences of using, so the SUD rate rises and reaches the child through $\delta_d$.

*Treatment First does not redistribute, removes an externality but creates two others.* Curing the parent is how the child's worse draw is avoided. But conditioning assistance means some households do not receive it and end up homeless, especially in the short run, and over that horizon it creates the two externalities Housing First removes. The short run matters on its own account: a household that wants to stop cannot do it immediately, and the transition is costly while it lasts, which is why the policy analysis reports the transition and not only the steady states.

*And general equilibrium returns part of it.* First, the drug price is endogenous, set by the supply curve that $\nu_1$ and $\nu_2$ describe, so curing disorders lowers drug demand, lowers the price, and makes using cheaper for those still deciding: part of what treatment removes comes back through lower prices. Second, when the government spends more on either policy and the tax rises, hours fall, the base shrinks and the tax has to rise again. Third, Treatment First has an offset of its own in the other direction: sending more households into treatment raises the recovery rate faced by everyone deciding whether to use, and those consequences affect a newborn when it makes its own decisions, so the policy changes behaviour beyond the households

it treats. And because $\kappa_x$ is levied on the aggregate headcount, any movement in homelessness feeds back into everyone's utility. So the mechanical effects above are upper bounds on both sides.

For every parameter the things worth knowing are what happens to the private value, to the externality and, where it applies, to the distortion, *as its value rises or falls*, and each entry says so in a line before giving the moment. The middle blocks are where the two move together: because the child inherits a higher chance of landing in exactly the states those parameters price, anything that raises the private cost of using or of being unhoused raises the externality as well. Both policies work on those blocks, from opposite ends.

Table 5: The twenty nine parameters that carry the story of this paper

| Parameter | Description | Set | Moment and source | Data | Model | Value |
|---|---|---|---|---|---|---|
| *Who uses drugs* | | | | | | |
| $\theta$ | Weight on drugs in utility | Int. | Drug spending over income, users, NSDUH 2024 | 35% | — | — |
| $b$ | Whether a household uses | Int. | Drug participation, NSDUH 2024 | 5.13% | — | — |
| $\lambda$ | SUD hazard per unit used | Int. | Adult SUD rate, NSDUH 2024 | 3.68% | — | — |
| $\bar{\kappa}_{\phi d}$ | Increased mortality with a SUD | Ext. | Mortality ratios, Hjorthøj et al. (2023) | 6 to 10 | n.a. | 8.00 |
| $\aleph$ | Flow value of being alive | Int. | Value of a statistical life, Viscusi and Aldy (2003) | \$7.5m | — | — |
| *What using and being homeless cost, and the voucher* | | | | | | |
| $\rho_d$ | Earnings lost to a SUD | Int. | Employment rate of the homeless, HUD 2023 | 50% | — | — |
| $\kappa_{ued}$ | Employment multiplier, SUD | Int. | Unemployment rate given a SUD, NSDUH 2024 | 16.25% | — | — |
| $h_1$ | Service flow of homelessness | Int. | Homelessness rate, HUD 2023 | 0.399% | — | — |
| $\kappa_{\phi H}(j)$ | Increased mortality when homeless | Ext. | Mortality ratios by age, Baggett et al. (2013) | 9.1 to 1.1 | n.a. | 4.41 to 1.09 |
| $\kappa_{ueH}$ | Employment multiplier, homeless | Ext. | Desmond and Gershenson (2016) | 0.80 | n.a. | 0.80 |
| $\zeta_{DI}$ | Disability replacement, SMI | Ext. | SSDI and SSI rates, SSA 2023 | 0.25 | n.a. | 0.25 |
| $\underline{c}$ | Food stamp consumption floor | Int. | Food stamps over labour income, USDA | 0.42% | — | — |
| $\pi_V$ | Voucher lottery probability | Int. | Recipients over eligible, HUD 2023 | 26.6% | — | — |
| *Mental illness chain* | | | | | | |
| $\iota$ | SUD multiplier on SMI onset | Ext. | Danish registers, Nielsen et al. (2017) | 2.00 | n.a. | 2.00 |
| $\lambda_m$ | Chance of recovering from SMI | Int. | SMI share of homeless, Gutwinski et al. (2021) | 25% | — | — |
| $\rho_m$ | Earnings lost to SMI | Int. | Labour income of the homeless relative to average | 19.1% | — | — |
| $\kappa_{uem}$ | Employment multiplier, SMI | Int. | Employment rate given SMI, NSDUH 2023 | 30% | — | — |
| $\mu_m$ | SUD recovery multiplier, SMI | Ext. | Dual diagnosis relapse, Bradizza et al. (2006) | 0.82 | n.a. | 0.82 |
| $\kappa_{\phi m}$ | Mortality multiplier, SMI | Ext. | Olfson et al. (2015) | 2.50 | n.a. | 2.50 |
| *Recovery and treatment* | | | | | | |
| $\bar{\pi}_r$ | Baseline SUD recovery rate | Ext. | Ten year remission rate, Vaillant (1995) | 0.025 | n.a. | 0.025 |
| $\mu_H$ | Recovery multiplier when homeless | Int. | Weeks abstinent, Milby et al. (2005) | 1.473 | — | — |
| $g$ | Treatment effect on recovery | Ext. | Odds ratio, Milby et al. (2005) | 1.35 | n.a. | 1.35 |
| $\bar{\tau}^u$ | Utility cost of treatment | Int. | Treatment take up, NSDUH 2023 | 24% | — | — |
| *Externalities and distortions* | | | | | | |
| $\omega$ | Weight on the child | Int. | Use rise with altruism off, NSDUH 2024 | 18.3% | — | — |
| $\delta_d$ | Child's earnings, parent SUD | Int. | Child SUD ratio, McGovern et al. (2023) | 3.00 | — | — |
| $\delta_h$ | Child's earnings, homeless | Int. | Child homelessness ratio, Bassuk et al. (1997) | 2.10 | — | — |
| $\kappa_x$ | Disamenity of homelessness | Ext. | Willingness to pay, Loubière et al. (2020) | 0.20% | n.a. | 0.002 |
| $\sigma_\ell$ | Curvature on leisure | Ext. | Frisch elasticity of 0.5, Chetty (2012) | 0.50 | n.a. | 3.00 |
| *Drug market* | | | | | | |
| $\nu_2$ | Drug supply elasticity | Ext. | Cannabis supply elasticity, Mace et al. (2020) | 1.46 | n.a. | 1.68 |

Note: n.a. in the Model column marks a parameter that is set from outside rather than calibrated inside the model. For those the model produces no internal target to compare with the data, so the cell reads n.a. rather than a dash, which keeps them distinct from the internally calibrated parameters.

**Who uses drugs.** These four set participation and, more importantly, *who* participates. Neither policy acts on them directly; they decide how much there is for a policy to act on.

**$\theta$, the weight on the drug in utility.** How much a household wants the drug at all, against consumption and housing. It is the level of the temptation where $b$ is its curvature, and it is the parameter the amount spent on drugs, when it does, turns on. As it rises use rises on the extensive and intensive margins, so the private cost and the externality both rise and there is more for either policy to act on. It is calibrated internally to drug spending over income across users, 35 percent in NSDUH 2024. The moment is built in two steps, because users are not one group. Households that buy without carrying a disorder and households that carry one have very different ratios, so each group's own total drug spending is divided by that group's own total income, and the two ratios are then averaged with their

shares of the using population as weights:

$$\text{drug share} \;=\; \frac{\mu_e}{\mu_e+\mu_d}\,\frac{p_v\bar{v}_e}{\bar{y}_e} \;+\; \frac{\mu_d}{\mu_e+\mu_d}\,\frac{p_v\bar{v}_d}{\bar{y}_d},$$

where $\bar{v}_g$ and $\bar{y}_g$ are the average drug quantity and average income in group $g$, and $\mu_e$ and $\mu_d$ are the masses without and with the disorder. The target follows from the two group ratios and the split between them: users without a disorder spend 11 percent of income on the drug and users with one spend 44, and the share of users carrying a disorder is the SUD rate over the participation rate, $0.0368/0.0513 = 0.717$, so the average is $0.717 \times 0.44 + 0.283 \times 0.11 = 0.35$.

**$b$, whether a household uses at all.** The marginal value of the drug goes as $\theta/(b+v)$, so a larger $b$ makes the first units worth less and a household with little to spend buys nothing. As $b$ rises use falls, so the private cost and the externality both fall, and housing assistance is less likely to be translated into more spending on drugs: $b$ is the size of the moral hazard. Its value is chosen so the model generates a drug participation rate of 5.13 percent, the NSDUH 2024 figure for adults aged 26 and over, which identifies it sharply because participation is close to a threshold function of $b$.

**$\lambda$, the hazard per unit used.** This is what makes using *more* risky, and it is the step that turns spending into a disorder. As $\lambda$ rises both the private cost and the externality rise, and Housing First's moral hazard costs more, since it is $\lambda$ that converts an extra voucher into a higher SUD rate Treatment First attempts to cure. Its value is chosen so the model generates the share of adults carrying a SUD, 3.68 percent among those aged 26 and over in NSDUH 2024.

**$\bar{\kappa}_{\phi d}$, increased mortality with a SUD.** This is the parameter the whole paper turns on. Mortality truncates the value of life for a user, which hits households with the highest life values hardest and thus reverses the income gradient, making the poor the users rather than the rich. As it rises the private cost rises, and the externality rises with it, since a child born to a parent with a SUD has a lower value of life. Both policies reduce these deaths, treatment by curing and housing because recovery is less likely when homeless. It is calibrated externally, at the midpoint of the standardised mortality ratios for SUD in Hjorthøj et al. (2023), which run six to ten, with the age profile from CDC WONDER 2023.

**$\aleph$, the flow value of being alive.** The other half of the mortality mechanism: $\bar{\kappa}_{\phi d}$ sets how much a SUD raises the chance of dying and $\aleph$ sets what the life being lost is worth. As it rises the private value of every life either policy saves rises, and use falls, so the externality falls. Its value is chosen so the model generates a value of a statistical life of \$7.5m, from Viscusi and Aldy (2003).[11]

**What using and being homeless cost, and the voucher.** Both policies work on this block, from opposite ends: each parameter says how bad a state is, so each sets what a voucher is worth to a household in it and what being cured of the reason for it is worth. The voucher itself closes the block, since it is the instrument that acts on all of them. Because the child inherits a higher chance of landing in these states, everything here moves the private value and the externality together.

**$\rho_d$, earnings lost to a SUD while still working.** It reduces the value of life when a household acquires a SUD, since the earnings it costs would have been enjoyed for the rest of that life. It also does a job nothing else can: it lets an *employed* household be poor enough to be priced out of housing, which an employment effect alone cannot deliver, since that makes essentially every homeless household jobless, whereas about half are not in the data. As $\rho_d$ rises the penalty shrinks, so the private cost falls and, because the impact on the child is less harmful, the externality declines. These earnings are lost whether the household is housed or homeless, so treatment affects them and housing does not.[12] Its value is chosen so the model generates the employment rate of the homeless, 50 percent, from HUD's 2023 point-in-time count and TEDS-A 2022.

[11] I check that no household has a negative value of life in equilibrium, so death is never an optimal outcome.

[12] One subtlety is worth stating, because it may seem like a contradiction with Section 3.1. There the proportional income penalty was shown to matter little for *who* uses. That is a statement about the calibrated model, not the parameter: raising $\rho_d$ with $b$ held fixed does change who uses, but $b$ is recalibrated to the participation rate each time the penalty changes, and once it is, the composition of users is back where it started.

$\kappa_{ued}$**, the employment multiplier for a SUD.** The same channel on the extensive margin: regaining employment is less likely and losing it more likely with a disorder. As it rises the private cost rises, even more than for $\rho_d$, the externality rises with the child's draw, and the tax base narrows since the household does not work and receives unemployment insurance, so it costs taxpayers as well. Its value is chosen so the model generates the unemployment rate given a SUD, 16.25 percent in NSDUH 2024.

$h_1$**, the service flow of homelessness.** It sets how bad the outside option is, and the value of being alive is measured against that outside option: a household with little to lose from dying is one for which the mortality cost of using is small. As $h_1$ rises the street is less bad, so the private cost of homelessness falls, a voucher is worth less, and the externality on a child likelier to be unhoused falls with it. Its value is chosen so the model generates the homelessness rate, 0.399 percent of households in HUD's 2023 point-in-time count.

$\kappa_{\phi H}(j)$**, mortality when homeless.** As it rises the private cost of being unhoused rises and thus so does the private value of a voucher. Baggett et al. (2013) give race adjusted mortality ratios for homeless adults of about 9.1 at ages 25 to 44, falling to 1.1 at 65 to 84. Those are totals for a population most of which also carries a SUD, so it is set externally age by age net of that composition, which is why it is derived jointly with $\bar{\kappa}_{\phi d}$ rather than read off directly, as set out in Section 4.8 at (25).

$\kappa_{ueH}$**, employment lost to homelessness.** Here the homelessness itself costs the job, so housing mechanically recovers the earnings and treating does not. As it rises the private cost and the externality rise together, which makes it the clearest case in the table of a parameter loading Housing First's side, and it also reduces the tax burden since these households are more likely to be employed. It is set externally at 0.80, from Desmond and Gershenson (2016).

$\zeta_{DI}$**, the disability transfer for the non employed with SMI.** It is the floor for one of the worst states in the model, being ill and out of work, so it sets how far a household can fall. As it rises the private cost of that state falls, and the externality on a child likelier to reach it falls, at the price of a higher tax on everybody else. It is set externally at 0.25, from SSDI and SSI replacement rates, SSA 2023.

$\underline{c}$**, the food stamp consumption floor.** The floor under consumption, and it belongs here rather than among the standard parameters. As it rises the private cost of every bad state falls, so a voucher is worth less at the margin and the externality on a child likelier to reach those states falls too, while the distortionary taxes that fund it rise. It is calibrated internally to food stamp spending over labour income, 0.42 percent, from the USDA.

$\pi_V$**, the chance of drawing a voucher.** This is the policy dial itself. As it rises the private gain rises and two externalities fall mechanically, the child's penalty for growing up unhoused and the aggregate disamenity, while the third rises through the SUD rate and the tax rises with the bill: the whole trade in one parameter. Its value is chosen so the model generates the 26.6 percent ratio of voucher recipients to the income eligible population in HUD's Picture of Subsidized Households, 2023.

**Mental illness chain.** This block is why Treatment First reaches further than the disorder it treats: a SUD raises the chance of mental illness, mental illness lowers earnings and employment and raises death, and the child is susceptible to all of this. Everything here therefore moves the private cost and the externality in the same direction, and the first three decide whether the chain exists at all.

$\iota$**, the SUD multiplier on SMI onset.** The hinge of the whole block. Without it mental illness is exogenous to everything a policy does and treating SUD cannot touch it. As it rises a SUD causes more mental illness, so the private cost of using rises, and the externality rises further, because the child of a user now inherits a higher chance of illness on top of the productivity penalty. Treatment First's return rises with them, because curing the disorder now prevents the illness too. It is set externally at two, from the Danish registers in Nielsen et al. (2017).

$\lambda_m$**, the annual chance of recovering from SMI.** The persistence of the state, and the one piece of the chain the data cannot hand over, since prevalence pins the stock and not the flows. As it rises mental illness is less persistent, so the private cost falls, because fewer years are spent ill, and the externality falls with it, since a child who is likelier to acquire a SUD and through it SMI has a higher value of life. It is calibrated internally to the SMI share of the homeless in Gutwinski et al. (2021), 25 percent.

$\rho_m$**, earnings lost to SMI.** The counterpart of $\rho_d$ for mental illness, and it reaches the comparison through $\iota$: a household carrying the disorder faces onset at $\iota$ times the baseline rate, so treatment reduces mental illness in a roundabout way and recovers these earnings for households that never fall ill. As it rises the penalty shrinks, so the private cost of illness falls and, because the child's draw improves with it, the externality falls too. Its value is chosen so the model generates the average labour income of the homeless relative to the average, 19.1 percent.

$\kappa_{uem}$**, the employment multiplier for SMI.** The same as $\rho_m$ but on the extensive margin, and where the chain does most of its work, since employment is what mental illness costs most. As it rises the private cost rises, because being ill now costs the job as well as the wage, and the externality rises with the child's draw; the tax base narrows, which is a distortion on top of both. A child of a treated parent is less likely to carry SMI, which raises its earnings and lowers its chance of being unhoused, so part of Treatment First's return never appears in the treated household's own drug use. Its value is chosen so the model generates the employment rate given SMI, about 30 percent, the midpoint of the range for high disability diagnoses in NSDUH 2023.

$\mu_m$**, the SUD recovery multiplier for SMI.** The second direction of the chain: not only does a SUD cause mental illness, mental illness means the SUD is less likely to be cured. As it falls the disorder is harder to leave, so the private cost of acquiring one rises and the externality rises with it, and treatment has more to overcome in exactly the households it most wants to reach. It is set externally at 0.82 from dual diagnosis relapse rates in Bradizza et al. (2006), and it is one of the three values in the paper carrying no real discipline.

$\kappa_{\phi m}$**, the mortality multiplier for SMI.** The chain's last link, and what makes it matter for welfare, not through income. As it rises deaths among the mentally ill rise, so the private cost rises, and the externality rises with it, because a child born to a parent with a SUD is more likely to acquire a SUD and through it SMI, and therefore more likely to die. This is how a SUD affects mortality indirectly, through SMI, rather than only by the direct route. It is set externally at 2.5, the middle of the standardised mortality ratios for SMI in Olfson et al. (2015).

**Recovery and treatment.** Both policies work on this block, since recovery is what Treatment First buys directly and what Housing First buys by housing people. The first two set how much recovery there is to buy and the last two how much of it a policy can reach.

$\bar{\pi}_r$**, the baseline SUD recovery rate.** The hazard of getting out of the disorder for a housed household, which is the level everything else in this block scales. As it rises disorders end sooner on their own, so the private cost of acquiring one falls and the externality falls with it, and there is less for either policy to buy. It is set externally from the ten year remission rate in Vaillant (1995), which gives an annual hazard of 0.025.

$\mu_H$**, SUD recovery when homeless.** How much less likely recovery from the disorder is for an unhoused household than a housed one. It matters more than the level $\bar{\pi}_r$ because it is a *gradient*: it is what a voucher buys beyond shelter. As it falls the gradient steepens, so housing buys more recovery: the private value of a voucher rises and the externality falls. Near one, assistance is a transfer and the moral hazard stands unopposed. Its value is chosen so the model generates the increase in recovery housing produced in Milby et al. (2005), a randomised trial on cocaine dependent homeless adults: 6.60 weeks abstinent in the housed arm against 4.48 homeless, so housing raises recovery by 47.3 percent.

$g$**, the treatment effect on recovery.** Treatment First's instrument, and the parameter its case rests on most directly. As it rises treatment works better, so a household that carries the disorder is more easily cured, and therefore the private cost falls, with the household spending less of its life carrying one. The externality falls with it, and by the same route: a child who acquires a disorder is also more easily cured, so what its parent hands on costs it less of its own life. It is set externally at 1.35, the odds ratio in Milby et al. (2005).

$\bar{\tau}^u$**, the utility cost of treatment.** Take up is a choice, not a rule, and this is what makes households decline treatment they might gain from in the future. As it rises fewer take treatment, so the externality rises, since more people remain with a SUD and thus more children are born with lower incomes, which leads to higher SUD incidence, and the private cost rises too, since treatment is costly. It is calibrated

internally to treatment take up among adults with a SUD, 24 percent in NSDUH 2023.

**Externalities and distortions.** This block is about what causes the externalities and the distortions. This is where the ranking between Housing First and Treatment First is decided, since the private comparison already favours Housing First, holding externalities and distortions aside. The first three are what a parent's disorder and a parent's homelessness cost the child, and the fourth is what the homelessness of strangers costs everybody. All four are externalities in the strict sense: one household's choice enters another household's utility directly without being internalised by the household making the decision, and it would still do so if the programmes were paid for with a non distortionary tax, such as a lump sum tax. The last is not an externality but a distortion, and it closes the block for that reason: it says what it costs to raise the revenue either programme needs rather than whose utility a choice lands in.

$\omega$**, the parent's weight on its child.** A parent bears $\omega$ of what its own state costs its child, and $\omega$ is small. As it rises the parent counts more of the child's loss, so the externality left for a policy to correct is smaller. Identifying it in the model requires care, because *every* household here has a child, born after it dies, so there is no parent against non parent comparison to run in the benchmark. The moment is a counterfactual instead. In NSDUH 2024 past month use among parents is 0.845 of use among non parents, with parents reweighted to the non parents' joint distribution of age, income, sex and marital status so the gap is behaviour rather than composition. The child term of the household's problem in Section 4.11, equation (30), is the only thing that makes a household in the model use less than it would if it did not have a child, so its value is chosen so that removing the child's penalty at the calibrated parameters raises drug use by 18.3 percent, which is that same ratio read the way the model reads it: the counterfactual raises use from the parent level.

$\delta_d$**, what the child loses to a parent's SUD.** The child does not inherit the disorder, only a worse productivity draw, and that lower expected draw is what makes it likelier to be poor, to use, to acquire a disorder and mental illness consequently, to be unhoused and to die earlier. As it rises the externality rises, and it loads whichever policy prevents more disorder: Treatment First by curing the parent, Housing First because recovery is easier once housed, while the parent's own private cost is largely unchanged, which is exactly what makes it an externality. No paper estimates the effect of a parent's SUD on a child's lifetime earnings directly, which is why the penalty is calibrated to the disorder ratio rather than lifted from an estimate. What the literature supplies is the two halves of the chain: Felitti et al. (1998) link household dysfunction to adult outcomes, and Chetty et al. (2016) measure what childhood environment does to adult earnings, against which the calibrated penalty is checked. It is calibrated inside the model to reproduce the ratio of disorder rates between the children of parents with and without a disorder, which McGovern et al. (2023) put between two and four; the midpoint, three, is the target. The moment does not need the parent's type carried through the child's whole life. The only thing $\delta_d$ changes about a child is the productivity it is born with, and the model already holds the distribution of that draw conditional on the parent's disorder, and the disorder rate at each productivity level; the ratio is those two contracted against each other, at the same parental housing status so the comparison isolates the disorder.

$\delta_h$**, what the child loses to homelessness.** It works through the same compounding draw as $\delta_d$, and as it rises it loads whichever policy houses more people: Housing First directly, Treatment First by removing the disorder that leads to homelessness in the first place, again leaving the private cost mostly unchanged. It is calibrated internally, on the transmission of its own condition, exactly as $\delta_d$ is: Bassuk et al. (1997) compare 220 homeless mothers with 216 low income housed mothers and find that 14.2 against 8.3 percent had a primary female caretaker who had been homeless, and 10.3 against 4.4 percent a male one, odds ratios of 1.80 and 2.41. $\delta_h$ is set so the child of a homeless parent is 2.1 times as likely to be homeless itself, the midpoint of the two. Because both arms of that study are poor, the target isolates transmission from the poverty that accompanies it. Chetty et al. (2016), who find that moving a child to a lower poverty neighbourhood before age thirteen raises adult earnings by 31 percent, is then an over identifying check rather than the source.

$\kappa_x$**, the disamenity of homelessness.** The only term through which one household's housing status enters the utility of another who is not its own child, which is what makes the homelessness rate a genuine fixed point problem rather than accounting. As it rises the externality from the aggregate headcount

grows, which loads whichever policy houses more people. Setting it high should mostly favour Housing First, so the higher figure is taken as the baseline and a ranking that still favours Treatment First survives the assumption least kind to it. It is calibrated externally from the contingent valuation in Loubière et al. (2020): €57 a year among the 42 percent of respondents willing to pay anything, which is 0.20 percent of average annual earnings, against €23 and 0.08 percent averaged over all respondents.

$\sigma_\ell$**, the curvature on leisure.** The curvature that turns $\omega_\ell$ into an elasticity: together they decide how far hours actually move when the after tax wage does. The two are not the same dial. $\omega_\ell$ sets what an hour of leisure is worth against consumption, so it fixes the level of hours; $\sigma_\ell$ sets how fast that worth falls as leisure is given up, so it fixes how far hours move when the wage changes. One pins the average, the other the response to a tax. It belongs here because the size of the distortion is exactly what this parameter sets: at a high curvature hours barely respond and the tax that funds either policy is close to a lump sum, at a low one the same revenue costs far more in forgone work. As it rises labour supply is less elastic, so the distortion falls and both policies look cheaper, without touching the private value or the externality at all. It is set externally to produce a Frisch elasticity of 0.5, from Chetty (2012).

**Drug market.** One parameter of the drug market belongs here. It is what makes the drug price endogenous, so drug use affects drug demand and the drug price in equilibrium; the normalisation and the price it fixes are standard and sit in Table 6. This is a general equilibrium offset that runs against Treatment First in particular. Curing households takes them out of the drug market, demand falls and so does the price, and a cheaper drug draws other households in. Some of what the policy removes is handed back through the price, and how much depends entirely on this parameter.

$\nu_2$**, the drug supply elasticity.** Set so that the price elasticity of supply is 1.46, which is what Mace et al. (2020) estimate for cannabis on Washington State administrative data. There is no direct estimate for an illicit drug, and cannabis is the closest market there is: it remains a Schedule I substance federally, so even a state licensed producer works without interstate trade, without ordinary banking and under federal prohibition, which is the set of frictions an illicit supplier faces. As it rises supply is more elastic, so the price moves less when demand does, and the general equilibrium offset to treatment shrinks: at a very elastic supply, curing households barely lowers the price, so general equilibrium effects play little role.

## 5.2 Standard parameters

The remaining twenty four parameters are more conventional, and I record them so that the calibration can be reproduced. Eight of them I calibrate inside the model, against a moment, because no source reads them off directly; the other sixteen I set from outside, from an estimate or from statute. Table 6 and the entries below run in the same seven blocks, and each entry gives the moment or the source that fixes the parameter together with the value that moment takes in the data. The Model column reports what the model produces against that value, and it does so only for the eight I calibrate internally, since the other sixteen have no target to reproduce.

These parameters are not inert in the experiments. For example, the discount factor governs how heavily a household weighs a death later in life, the dispersion of permanent income determines how many households are poor enough to be at risk of homelessness and of acquiring a disorder, and risk aversion governs what insurance is worth. What separates them from the parameters above is not that they do nothing, but that they are not specific to this paper.

**Preferences.** The four parameters in this block say what a household wants, and what it is willing to give up to get it.

$\sigma$**, risk aversion.** Set to the standard value of two, following Attanasio and Weber (2010). As it rises consumption is smoother and insurance is worth more, so the voucher is worth more; it also sets the marginal utility of consumption, so it enters every welfare comparison in the paper.

$\gamma$**, the weight on housing services.** The Cobb-Douglas share of the bundle that goes on housing, so it fixes what a dwelling is worth against consumption and against the drug. I choose it so that housing

Table 6: Standard parameters

| Parameter | Description | Set | Moment and source | Data | Model | Value |
|---|---|---|---|---|---|---|
| *Preferences* | | | | | | |
| $\sigma$ | Risk aversion | Ext. | Attanasio and Weber (2010) | 2.00 | n.a. | 2.00 |
| $\gamma$ | CD weight on housing | Int. | Housing over expenditure net of drugs, CEX 2024 | 26.8% | — | — |
| $\beta$ | Discount factor | Int. | Wealth over output, BEA fixed assets | 3.00 | — | — |
| $\omega_\ell$ | Weight on leisure | Int. | Average hours worked, Guvenen et al. (2023) | 0.40 | — | — |
| *Rental ladder* | | | | | | |
| $\underline{h}$ | Smallest rental size | Int. | Share in smallest unit size, ACS 2023 | 13.2% | — | — |
| $\bar{h}$ | Largest rental size | Int. | Share in largest unit size, ACS 2023 | 62.7% | — | — |
| $\varsigma$ | Curvature of the ladder | Ext. | Set; the two ends carry the distribution | 1.80 | n.a. | 1.80 |
| *Earnings and employment* | | | | | | |
| $\varepsilon(j)$ | Age profile of efficiency | Ext. | Hump peaking at age 56, Greenwood et al. (2025) | profile | n.a. | profile |
| $\sigma_y$ | Dispersion of the fixed draw | Int. | Bottom half's share of earnings, SCF 2022 | 17.2% | — | — |
| $\bar{\pi}^{eu}$ | Job separation rate | Int. | Unemployment rate, BLS CPS 2014–24 | 4.85% | — | — |
| $\bar{\pi}^{ue}$ | Job finding rate, annual | Ext. | Krueger et al. (2014) | 0.55 | n.a. | 0.55 |
| *Survival and mental illness* | | | | | | |
| $\bar{\phi}(j)$ | Baseline survival by age | Ext. | SSA period life table, ages 26 to 85 | profile | n.a. | profile |
| $p_m(j)$ | Age profile of SMI prevalence | Ext. | Share with SMI, NSDUH 2023 | 5.5% | n.a. | 5.5% |
| *Interest, taxes and public spending* | | | | | | |
| $r$ | Risk free rate | Ext. | Kaplan et al. (2020) | 0.03 | n.a. | 0.03 |
| $\xi$ | Tax progressivity | Ext. | Heathcote et al. (2017) | 0.85 | n.a. | 0.85 |
| $\tau_k$ | Capital income tax | Ext. | US long run rate on gains and dividends | 0.15 | n.a. | 0.15 |
| $\tau_\ell$ | Tax level | Int. | Average labour income tax, McDaniel (2007) | 22.4% | — | — |
| *Transfers and the voucher rule* | | | | | | |
| $\psi$ | UI replacement rate | Ext. | DOL statutory replacement rates | 0.50 | n.a. | 0.50 |
| $\bar{y}$ | Cap on insurance earnings | Ext. | DOL state benefit maxima | 1.00 | n.a. | 1.00 |
| $\kappa_V$ | Voucher eligibility cutoff | Ext. | HUD rule | 0.50 | n.a. | 0.50 |
| $\kappa_r$ | Tenant contribution rate | Ext. | HUD rule | 0.30 | n.a. | 0.30 |
| *Drug market* | | | | | | |
| $J_E$ | Last experimentation age | Ext. | Model age 55, that is age 80 | 55 | n.a. | 55 |
| $\nu_1$ | Drug supply shifter | Ext. | Normalisation, so $p_v = 1$ | norm. | n.a. | norm. |
| $p_v$ | Price of drugs | Ext. | Clears the drug market in the benchmark | 1.00 | n.a. | 1.00 |

Note: n.a. in the Model column marks a parameter that is set from outside rather than calibrated inside the model. For those the model produces no internal target to compare with the data, so the cell reads n.a. rather than a dash, which keeps them distinct from the internally calibrated parameters.

takes 26.8 percent of expenditure net of drugs, as shelter plus utilities do in the 2024 Consumer Expenditure Survey, which is the same moment and the same definition the stylized model of Section 3 uses. As $\gamma$ rises housing is worth more against consumption and the drug, households buy more of it, and that expenditure share rises with it.

**$\beta$, the discount factor.** It is chosen so that wealth over output matches the national accounts, a ratio of three, from the BEA fixed asset tables against NIPA output. As it rises households are more patient, save more, and the wealth to output ratio rises with it.

**$\omega_\ell$, the weight on leisure.** It is set so that the model generates average hours worked of 40 percent, from Guvenen et al. (2023). As it rises leisure is worth more, households work less, and average hours fall with it.

**Rental ladder.** These three parameters build the grid of dwelling sizes a renter chooses from, and they fix how the rungs are spaced.

**$\underline{h}$ and $\bar{h}$, the rental ladder.** The smallest and the largest rental size. The ladder is pinned at both ends and the rung between them follows from the curvature $\varsigma$. I choose the two ends to match the two ends of the distribution of occupied units by number of bedrooms in the 2023 American Community Survey, table B25042: 13.2 percent of households sit in the smallest category and 62.7 percent in the largest. Three shares carry two free numbers, so the two sizes identify them and the middle share follows by complementarity, which is why I set the curvature rather than calibrate it. Each end moves its own share, which is what identifies them separately. The universe is occupied units, so the model counterpart conditions on being housed, and the homeless are in neither denominator.

**$\varsigma$, the curvature of the rental ladder.** I set it at 1.8 rather than calibrate it, because with three rental

sizes the size distribution carries two free numbers and the two ends of the ladder already take them. What the curvature does is place the middle rung: the middle size sits a fraction $(1/2)^{\varsigma}$ of the way from $\underline{h}$ to $\bar{h}$, so at $\varsigma = 1.8$ it sits 28.7 percent of the way up from the smallest rental rather than the 50 percent a uniform ladder would give. At the calibrated ends the three rental sizes stand in the ratio $1 : 1.46 : 2.59$. The two renter cost shares are left untargeted and serve as a check instead: in the 2023 American Community Survey 49.5 percent of renter households spend more than thirty percent of income on housing and 26.5 percent spend more than half, and what the model produces against those two numbers is reported with the other untargeted moments in Section 5.3.

**Earnings and employment.** These five parameters determine what a household is expected to earn over its life, how much of that is permanent, and how it moves between work and unemployment.

$\varepsilon(j)$**, the age profile of efficiency.** The deterministic hump in earnings over the life cycle, taken as the exponential of a quadratic in age used by Greenwood et al. (2025), $\varepsilon(j) = \exp\{-(j-1)^2/1800 + (j-1)/30\}$ with $j = 1$ at age 26. It peaks at age 56, having risen 65 percent from age 26, and then falls 4 percent from the peak to age 65, the last working age. Without it every household would earn the same at every age, and the ages at which households are poor, and so at risk of homelessness and of using, would be wrong at both ends of working life.

$\sigma_y$**, the dispersion of productivity.** Productivity is permanent: a household draws it once and keeps it for life, exactly as in the stylized model, so all that is needed is the spread of that draw. It is calibrated, to the share of earnings held by the bottom of the income distribution among working households. It is what decides how many households are poor enough to be at risk of homelessness and of acquiring a SUD. In the 2022 Survey of Consumer Finances, households aged 26 to 65 with positive wage and salary income hold 17.2 percent of total earnings in the poorer half of the distribution, and $\sigma_y$ is chosen to reproduce that share. The share falls in $\sigma_y$, so the mapping is monotone and one number identifies it. Two untargeted shares come out alongside: the model's top tenth holds 38.5 percent of earnings against 41.1 in the data, and its bottom tenth holds 1.1 percent against 0.99.

$\bar{\pi}^{eu}$**, the job separation rate.** Chosen to reproduce the aggregate unemployment rate in the BLS Current Population Survey (CPS) over 2014 to 2024, 4.85 percent, given the job finding rate set externally below. As it rises more households lose work each year, so the unemployment rate rises with it.

$\bar{\pi}^{ue}$**, the annual job finding rate.** The probability that an unemployed household finds work by the end of a period, which I set to 0.55 from the annual exit rate Krueger et al. (2014) estimate on matched CPS panels.

**Survival and mental illness.** These two parameters are age profiles that I take from outside the model rather than generate inside it.

$\bar{\phi}(j)$**, baseline survival by age.** The most recent Social Security Administration (SSA) period life table, table 4.C6, for ages 26 to 85, with the sexes combined by survivor weight rather than averaged. This is the profile every mortality multiplier in Section 5 multiplies, so it is the base the SUD, homelessness and mental illness hazards are measured against.

$p_m(j)$**, the age profile of SMI prevalence.** The target is the share of adults carrying serious mental illness at each age $j$ in NSDUH 2023, which integrates to 5.5 percent across all ages; the 5.5 percent is the overall level and the profile is what is matched, age by age. It sits here rather than in Section 5 because it is the baseline profile, in the same way the survival table is: what this paper adds is $\iota$, the multiplier a SUD puts on that onset, and that is where the interaction lives.

**Interest, taxes and public spending.**

$r$**, the risk free rate.** I set it to three percent, from Kaplan et al. (2020).

$\xi$**, tax progressivity.** The exponent on income in the tax schedule of Section 4.3, which is set to 0.85, from Heathcote et al. (2017). It is what makes the labour income tax progressive rather than flat. That in turn decides how much a programme's bill falls on the households it is meant to help, and how much on everybody else.

$\tau_k$, **the capital income tax.** I set it to fifteen percent, consistent with the statutory United States rate on long term capital gains and qualified dividends for the middle of the income distribution, which the Internal Revenue Service (IRS) sets out in Topic 409 (`https://www.irs.gov/taxtopics/tc409`).

$\tau_\ell$, **the level of the labour income tax.** It is calibrated so that the average labour income tax rate the model produces is 22.4 percent, which is what McDaniel (2007) measures for the United States and what Guvenen et al. (2023) use. It then moves with the government budget in every counterfactual, which is what makes both programmes paid for rather than free, and what makes the distortion of Section 5 a cost the model actually bears.

**Transfers and the voucher rule.**

$\psi$ **and** $\bar{y}$**, unemployment insurance.** These two set the replacement rate and the cap on the earnings it is paid on. I set them to fifty percent and one times average income, respectively, taking both from the statutory replacement rates and state benefit maxima that the Department of Labor compiles in its comparison of state unemployment insurance laws (`https://oui.doleta.gov/unemploy/statelaws.asp`).

$\kappa_V$ **and** $\kappa_r$**, the voucher rules.** These two set the income cutoff for eligibility and the tenant's contribution, and I take both from HUD's own rules rather than from any estimate. I set them to half of median income and thirty percent of income, respectively. The first is the very low income limit HUD publishes annually with its income limits (`https://www.huduser.gov/portal/datasets/il.html`); the second is the share a tenant pays of the rent when a voucher covers the rest, under the Housing Choice Voucher programme (`https://www.hud.gov/helping-americans/housing-choice-vouchers`).

**Drug market.** These three parameters describe the drug market: the first fixes who can enter it, and the other two fix the units in which its price and quantity are measured.

$J_E$**, the maximum age at which use can begin.** I set it to fifty five in model terms, which corresponds to age 80 in the data. It is there so that use cannot begin in the last few years of life, where almost no continuation value remains to be lost and the private cost of using is therefore close to zero. That is an artefact of the horizon rather than a feature of the world: to keep the age dimension manageable the model imposes certain mortality by age 85, so a household near the end knows death is coming with certainty and a disorder costs it almost nothing. Without the cap the model would let use begin exactly there, and at the final age it would be worse than that, since a household that acquires the disorder in the last period dies at the end of it whatever the hazard does: the disorder carries literally no consequence and only the flow benefit of using, which cannot be a reasonable description of the choice either.

$\nu_1$**, the supply shifter.** A pure normalisation, chosen so that supply equals demand at $p_v = 1$ in the benchmark. It normalises units, not behaviour.

$p_v$**, the drug price.** Set to one by normalisation in the benchmark, but free in every counterfactual. It is an equilibrium object rather than a dial.

## 5.3 Validation, and what is left undisciplined

None of the moments below is fitted. Each is produced endogenously by a model with sixty single ages, and each is a place the model can be caught. Section 9 of `data/DATA_APPENDIX.md` in the replication package derives the data side of every one of them, and it is the place to go for everything needed to replicate them: the dataset, the year, the universe, the computation, and the three cases where the universe rather than the arithmetic decides the number. The package ships those datasets as well, so none of them has to be downloaded first. The standardised mortality ratio below is age weighted. The last row is the intergenerational transmission the paper turns on, and it is the one moment here that needs a word about its construction. Bassuk et al. (1997) is a case control study of 220 homeless mothers and 216 low income housed mothers, so both arms are poor and the comparison isolates transmission from the poverty that accompanies it. The share whose primary female caretaker had ever been homeless was 14.2 against 8.3 percent, and for

the male caretaker 10.3 against 4.4, which are odds ratios of 1.8 and 2.4. A ratio taken against the general population instead would be far larger and would credit the housing penalty with everything else that differs between poor children, which Cobb-Clark and Zhu (2015) show is substantial: among households all on income support, those first homeless as children were more than twice as likely to have been in foster care.

| Moment | Data | Source |
|---|---|---|
| Homelessness rate, 25–34, 35–44, 45–54, 55–64, 65+ | 0.316, 0.338, 0.291, 0.250, 0.077 percent | HUD PIT 2024 |
| Shares of the homeless across those bands | 26.0, 27.2, 21.0, 18.4, 7.5 percent | HUD PIT 2024 |
| Returns to homelessness within 24 months | 17.4 percent | HUD SPM 2024 |
| Drug poisonings as a share of deaths, 25 to 44 | about 20 percent | CDC WONDER |
| Earnings shares, bottom tenth and top tenth | 0.99 and 41.1 percent | SCF 2022 |
| Bottom half's share of net worth | 2.52 percent | Fed DFA |
| Housing costs above 30 and 50 percent of income | 40.6 and 17.5 percent | SCF 2022 |
| Standardised mortality ratio, homeless | 9.1, 4.5, 1.1 by band | Baggett et al. |
| Tenant contribution rate | 31.2 percent | HUD |
| Voucher holders with wages as main income | 28 percent | HUD |
| Income, moderate or severe SUD over average | 71.1 percent | NSDUH 2024 |
| Labour force participation, SUD against clean | 65.8 against 61.5 percent | NSDUH 2024 |
| Homeless at treatment admission | 14.5 percent | TEDS-A 2022 |
| Psychiatric problem alongside SUD at admission | 43.1 percent | TEDS-A 2022 |
| Working at all, homeless in treatment | 6.6 percent | TEDS-A 2022 |
| Drug use disorder share of the homeless | 21.7 percent | Gutwinski et al. (2021) |
| Drug price participation elasticity | $-0.63$ | Saffer and Chaloupka (1999) |
| Odds a homeless adult had a homeless caretaker | 1.8 to 2.4 | Bassuk et al. (1997) |

Three parameters remain undisciplined, and they are flagged rather than hidden: the treatment effect on recovery $g$, the recovery multiplier for SMI $\mu_m$, and the drug supply curvature $\nu_2$. Of these only $g$ carries a headline result, and it is reported as a range rather than as a number.

# 6 The welfare criterion

Every welfare number in this paper is a consumption equivalent variation, and the convention matters enough to state exactly rather than leave a reader to infer. Write $V^b$ for lifetime value under the benchmark and $V^c$ for lifetime value under a counterfactual policy. The variation $\Delta$ is the proportional change in consumption, applied at every date and in every state, that makes a household under the benchmark as well off as the same household under the counterfactual:

$$V^b\big((1+\Delta)\,c\big) \;=\; V^c(c). \tag{40}$$

A positive $\Delta$ therefore means the counterfactual is worth that much of a permanent consumption gain, and a negative one that it costs that much. Reporting the compensating direction rather than the equivalent one is the reason a number can exceed one hundred percent: it is what consumption would have to rise *by*, not a share of anything.

**Whose welfare.** The household is a newborn, and the value is taken before any of its draws are realised, so $\Delta$ is an expectation over the productivity draw and everything that follows from it. This is the criterion behind the veil, and it is used rather than a utilitarian sum over the stationary distribution for a reason worth admitting: the two do not agree here, and the newborn criterion internalises the child by construction, which is part of what the paper is trying to establish. Both are therefore reported wherever they differ, with the newborn one primary and the utilitarian one beside it, so a reader can see which conclusion rests on which.

Two horizons are used, and they are the same object measured at different dates: a newborn on the date the policy is implemented, who lives through the transition, and a newborn in the new steady state. Because both are the ex ante value of a household at age one, any difference between them is the transition itself rather than a change in who is being counted. Averaging over the households alive at the reform would not have that property: that population is a mix of ages and states whose composition responds to the policy, so a channel could appear to grow with the horizon when what changed was the mix of people in the average.

**The stylized model has a closed form and the quantitative model does not.** With log utility over consumption and housing, scaling consumption by $1+\Delta$ at every date adds $K\log(1+\Delta)$ to lifetime value, where $K$ collects the weight on consumption and the discounted periods a household expects to live, $K=(1-\gamma-\theta)(1+\beta\phi_0)$. Equation (40) then inverts directly to $\Delta=\exp\{(V^c-V^b)/K\}-1$, which is how every consumption equivalent in Section 3.6 is computed.

No such inversion exists in the quantitative model, and three features of flow utility in equation (14) are why. The flow value of being alive $\aleph$ is additive, so it does not scale with consumption. The drug enters through $\log(b+v)$ with a Stone-Geary shifter, so scaling consumption does not scale the drug term proportionately. And the food stamp floor $\underline{c}$ binds for exactly the households the paper is about, so their consumption does not scale at all until the scaling lifts them off it. $\Delta$ is therefore solved numerically, by bisection on the newborn's expected value with the whole model re-solved at each trial, which is expensive but avoids assuming away the non-homotheticity that the mechanism depends on.

**What is scaled, and what is not.** The household's own consumption is scaled at every date and in every state. Two terms deliberately are not. The aggregate disamenity $\kappa_x H$ is additive in flow utility and does not scale with consumption, so it enters the level of welfare without entering the scaling; that is what lets it be switched off as a rung of the decomposition without disturbing the rest of it. The child term is a difference of newborn values, and the scaling is applied inside it consistently with the parent's, so a policy that improves the child's prospects is credited through the same measuring rod. Whether the child's own consumption is scaled as well is a choice rather than a fact, it moves the number, and the reported figures scale the parent only, with the alternative given alongside.

# 7 Experiments

*Plan for this section, not its content.* Four steps, set out in full in `code/EXPERIMENT_PLAN.md` and summarised here so the paper's structure is legible:

1. *The optimal generosity of each policy, separately*, over eligibility, the share of rent covered, and the probability of receipt, the last varied continuously so the question of whether the corner is optimal can be answered rather than assumed.
2. *Each optimum against the other policy at the same generosity*, which is what isolates conditionality from generosity. Both optima, because the optimal Housing First design is not the optimal Treatment First design. Reported under both readings of "the same": same design, and same cost.
3. *Transition against the long run*, for whichever policy wins, because a trial that runs six to twenty four months sees the first periods of a transition and not a steady state.
4. *The channel shutdown ladder at the optimum*, in both horizons: the child externality, the aggregate externality, the labour supply response that carries the tax distortion, the drug price response, and the cost of administering the test. The rungs do not commute, so a fixed order is reported together with the interaction term.

# 8 Discussion

*Plan for this section, not its content.* Four things to say once the numbers exist:

- *What the trials can and cannot settle.* If the ladder shows the externality nearly absent at implementation and growing with the horizon while the tax distortion bites immediately, that is the reconciliation: the trials are well powered for the margin that moves first and silent on the one that decides the ranking.
- *Which channel carries the result.* The paper insists on separating an externality from the distortions, and the ladder is what makes that separation a number rather than a claim. If the aggregate disamenity turns out to be small next to the intergenerational one, saying so is part of the point.

- *What would overturn it.* The three undisciplined parameters above, and the value of $\omega$, are where a reader should push. Report what each would have to be for the ranking to flip.
- *What the model does not have.* Neighbourhood effects on the child are priced through $\delta_h$ rather than modelled spatially; treatment capacity is not rationed; and the detection rates are set rather than estimated.

# 9 Conclusion

*Plan for this section, not its content.* Short. The moral hazard the debate argues about is real and is not an inefficiency, so it cannot carry the case for conditionality on its own. The case has to rest on an externality, on the distortions that fund the programme, or on the programme's own distortions, and those are three different claims. This paper prices the first and finds it intergenerational: what a parent's disorder and homelessness cost the child who replaces it. Close on what that implies for a policy debate currently being settled by executive order.

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